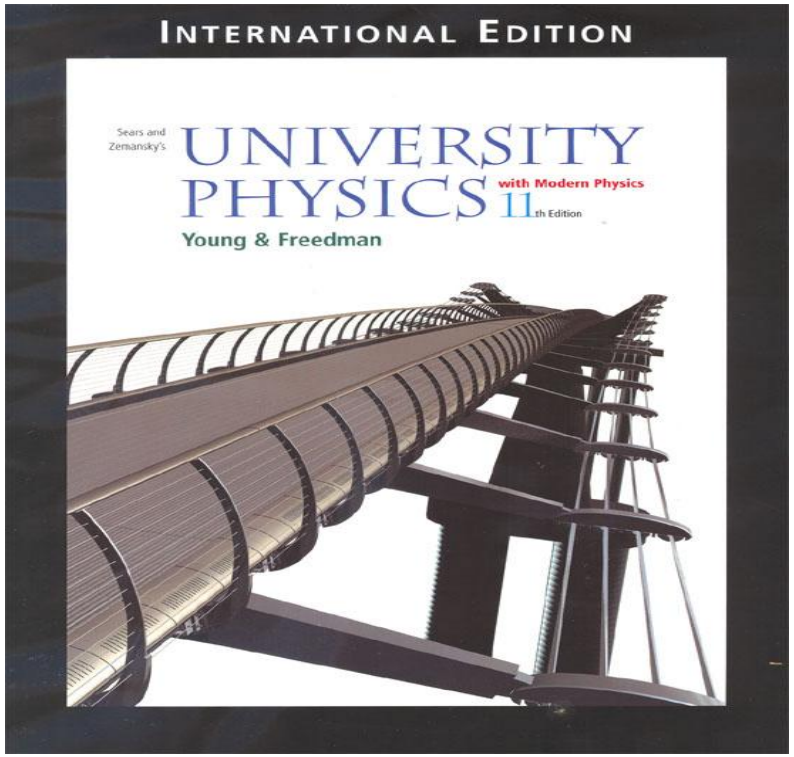
ELECTROMAGNETISM



21. Electric Charge and Electric Field

22. Gauss's Law

23. Electric Potential

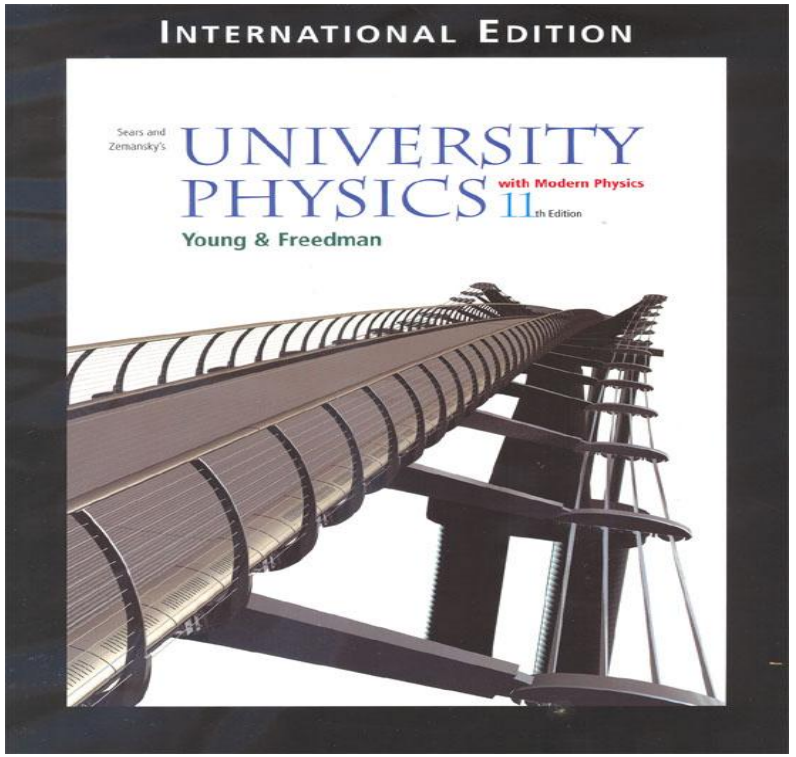
24. Capacitance and Dielectrics

25. Current, Resistance, and Electromotive Force

26. Direct-Current Circuits

ELECTROMAGNETISM

27. Magnetic Field and Magnetic Forces
28. Sources of Magnetic Field
29. Electromagnetic Induction
30. Inductance
31. Alternating Current
32. Electromagnetic Waves



Chapter 21

Electric Charge And Electric Field

Chapter Objectives

- The nature of electric charge
- Interactions of electric charges
- Coulomb's law
- The concept of electric field

Chapter Outline

1. Electric Charge
2. Conductors, Insulators, and Induced Charges
3. Coulomb's Law
4. Electric Field and Electric Forces
5. Electric-Field Calculations
6. Electric Field Lines
7. Electric Dipoles

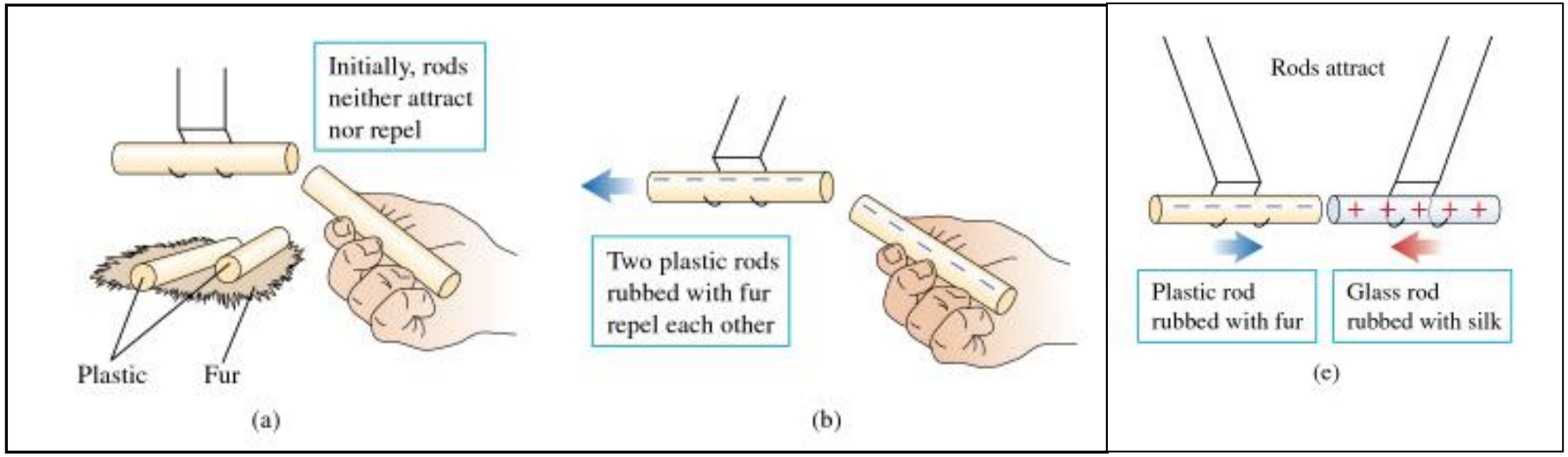
Chapter 21.1

Electric Charge

21.1 Electric Charge

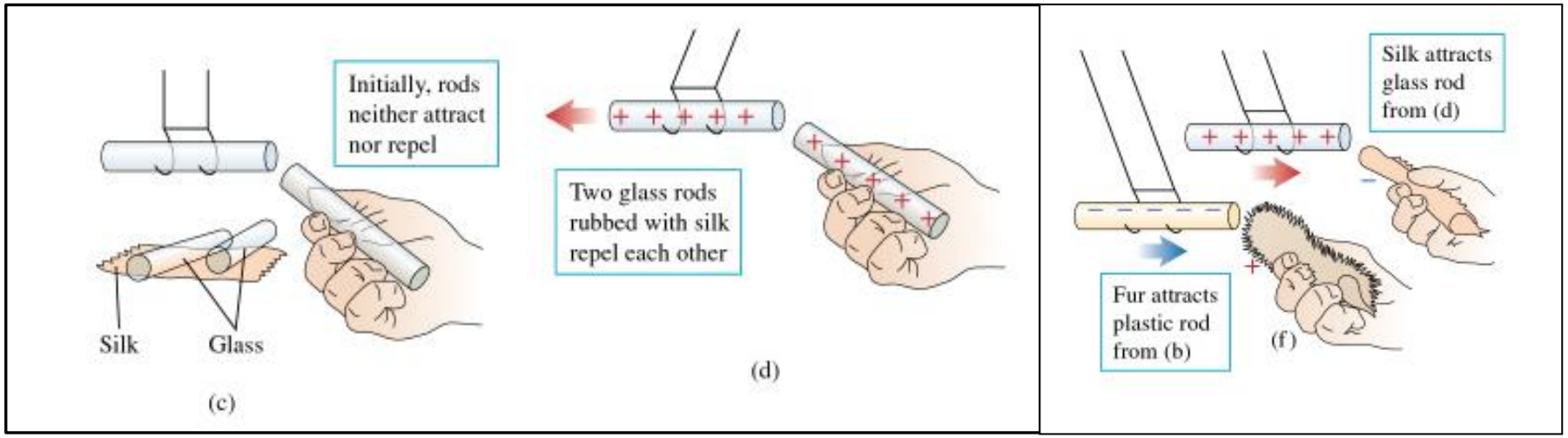
- Electric charge is an attribute that is as fundamental as mass.
- Electrostatics are defined as the interactions between electric charges that are at rest (or nearly so).
- The figures show some experiments used to demonstrate electrostatics.

21.1 Electric Charge



- Two negative charges repel each other. A positive charge and a negative charge attract each other.

21.1 Electric Charge



- Two positive charges repel each other. A positive charge and a negative charge attract each other.

21.1 Electric Charge

- Types of Charge

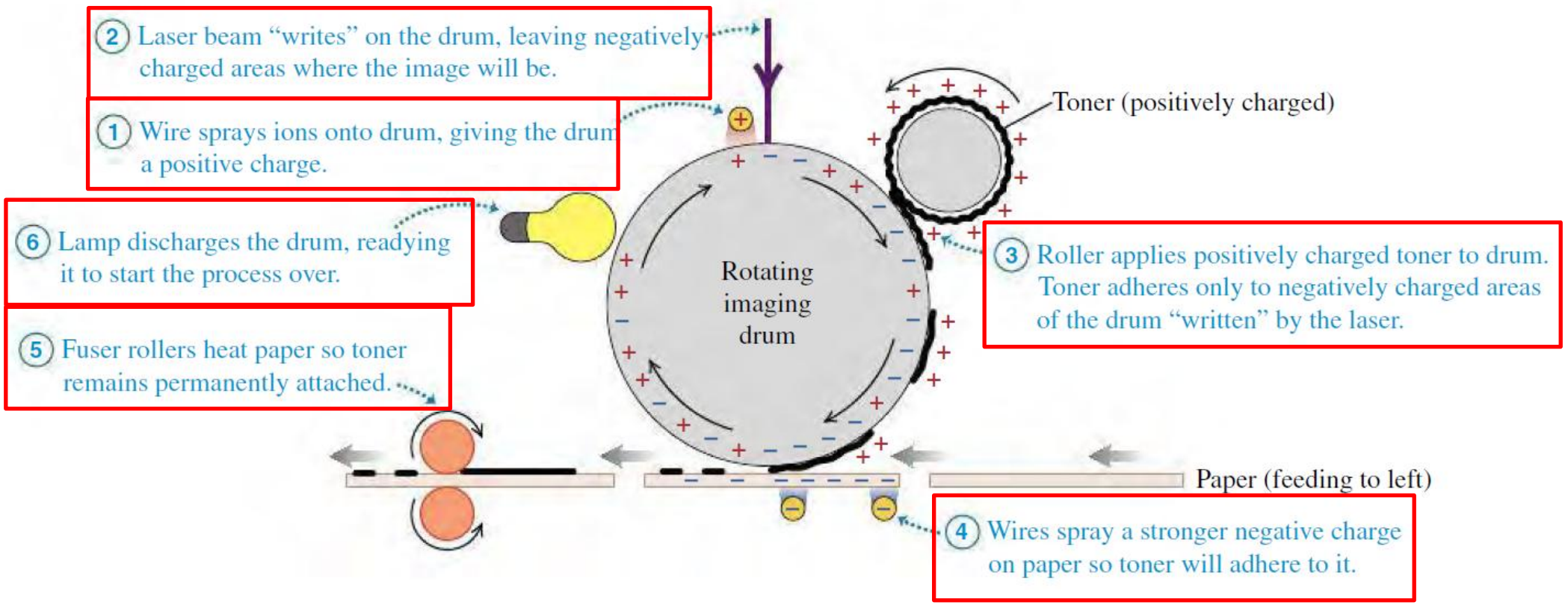
  There are exactly two kinds of electric charge, negative (-) and positive (+).

- Two positive charges or two negative charges repel each other. A positive charge and a negative charge attract each other.



- "Like charges" only mean that two charges have the same algebraic sign (both positive or both negative).
- "Opposite charges" means that the electric charges on both objects have different signs (one positive and the other negative).

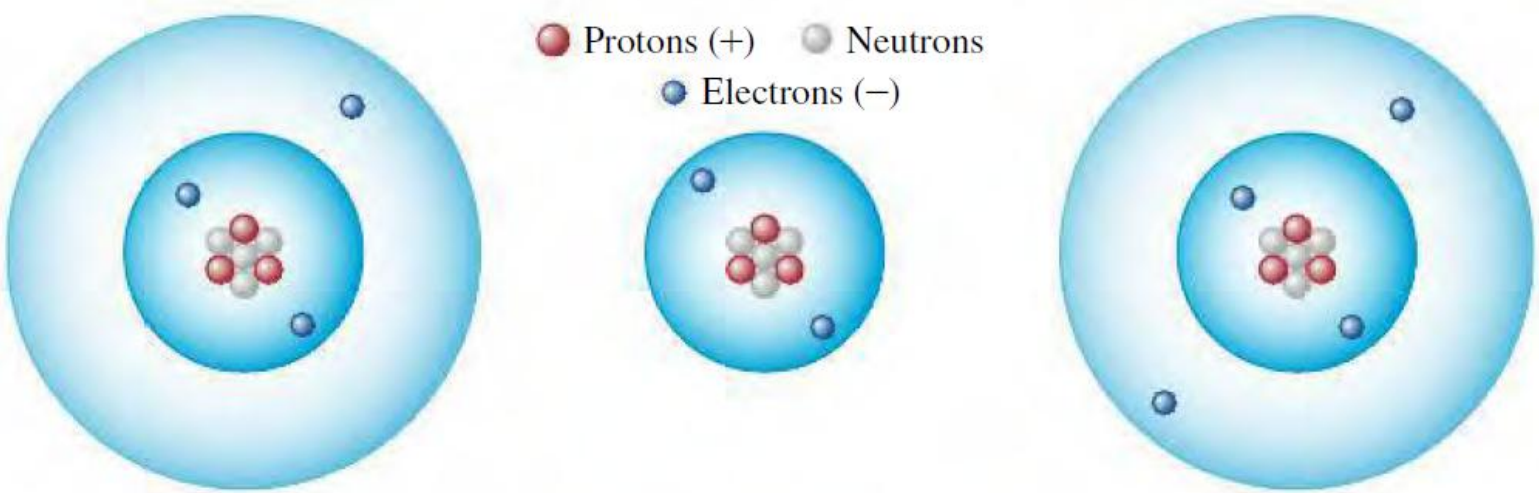
A technological application is a laser printer; the figure shows a schematic diagram of the operation of such a printer.


2 Laser beam "writes" on the drum, leaving negatively charged areas where the image will be.
1 Wire sprays ions onto drum, giving the drum a positive charge.
6 Lamp discharges the drum, readying it to start the process over.
5 Fuser rollers heat paper so toner remains permanently attached.
Toner (positively charged)
3 Roller applies positively charged toner to drum. Toner adheres only to negatively charged areas of the drum "written" by the laser.
Rotating imaging drum
Paper (feeding to left)
4 Wires spray a stronger negative charge on paper so toner will adhere to it.

Electric Charge and the Structure of Matter

- The atomic structure consists of three particles: the negatively charged electron, the positively charged proton, and the uncharged neutron.
- Protons and neutrons make up the nucleus while electrons orbit it from a distance.
- The figure shows how changes in the atomic structure of lithium determines its net electric charge.

Electric Charge and the Structure of Matter



(a) Neutral lithium atom (Li):	(b) Positive lithium ion (Li^+):	(c) Negative lithium ion (Li^-):
3 protons (3+)	3 protons (3+)	3 protons (3+)
4 neutrons	4 neutrons	4 neutrons
3 electrons (3−)	2 electrons (2−)	4 electrons (4−)
Electrons equal protons: Zero net charge	Fewer electrons than protons: Positive net charge	More electrons than protons: Negative net charge

Electric Charge and the Structure of Matter

- Atomic number is defined as the number of protons or electrons in a neutral atom of an element.
- A positive ion is formed by removing one or more electrons from an atom; a negative ion is one that has gained one or more electrons. This process is called ionization.
- When the total number of protons equals the total number of electrons in a macroscopic body, its total charge is zero and the body as a whole is electrically neutral.

Electric Charge and the Structure of Matter

- When we speak of the charge of a body, we always mean its net charge.

Conservation of Charge

- The Principle of Conservation of Charge: The algebraic sum of all the electric charges in any closed system is constant. (The net charge of a closed system never changes)
- In any charging process, charge is not created or destroyed but merely transferred from one body to another.
- The net charge: the algebraic sum of all the electric charges

Elementary Charge

- Definition: the magnitude of charge of the electron or proton is a natural unit of charge.
- Every observable amount of electric charge on any macroscopic body is always either zero or an integer multiple (positive or negative) of this basic unit, the electron charge – quantization of charge.
- The most fundamental unit of charge is the magnitude of the charge of an electron or proton, denoted by e, where

$$e = 1.602 \times 10^{-19}\ C$$

Chapter 21.2

Conductors, Insulators, and Induced Charges

21.2 Conductors, Insulators, and Induced Charges

- **Electric Conductors and Insulators**

  - **Conductors** of electricity are materials that permit electric charge to move easily through them;

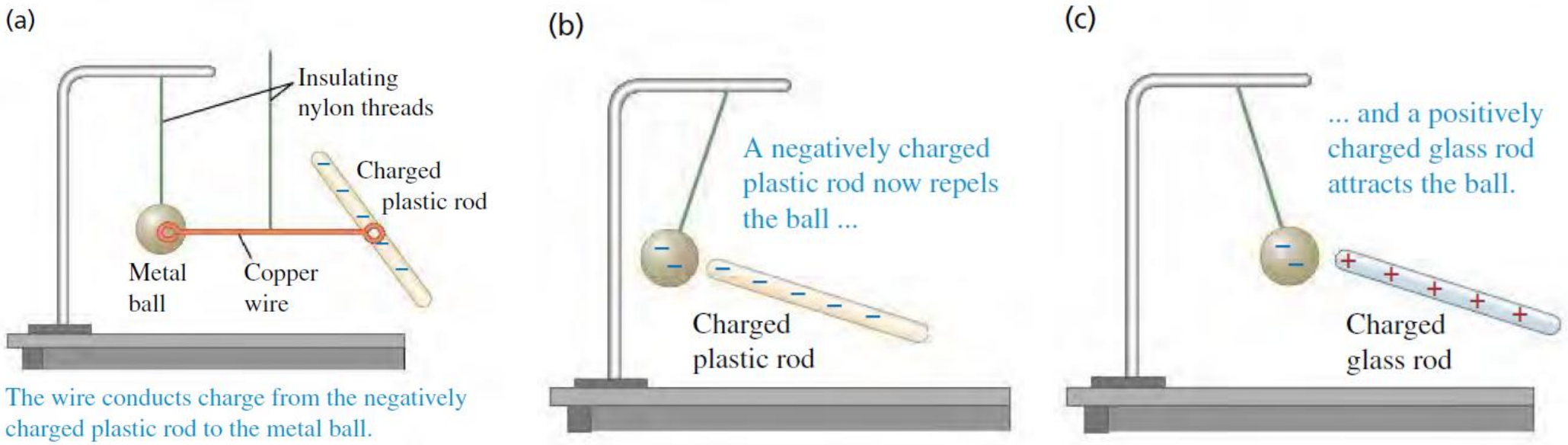
  - **Insulators** of electricity are material that do not permit electric charge to move easily through them.

    - ✓ Most metals are good conductors while most non-metals are insulators.

  - **Semiconductors** are intermediate in their properties between good conductors and good insulators.

21.2 Conductors, Insulators, and Induced Charges

The figure shows the use of copper as a good conductor, and glass and nylon as good insulators.



The wire conducts charge from the negatively charged plastic rod to the metal ball.

Ways to Charge

Charging Insulators by Rubbing

- If both objects had zero net charge before they were rubbed together, they now have net charges of equal magnitude and opposite signs, since charge is conserved.
- Two insulators are rubbed together to separate charge.

Charging a Conductor by Contact

Ways to Charge

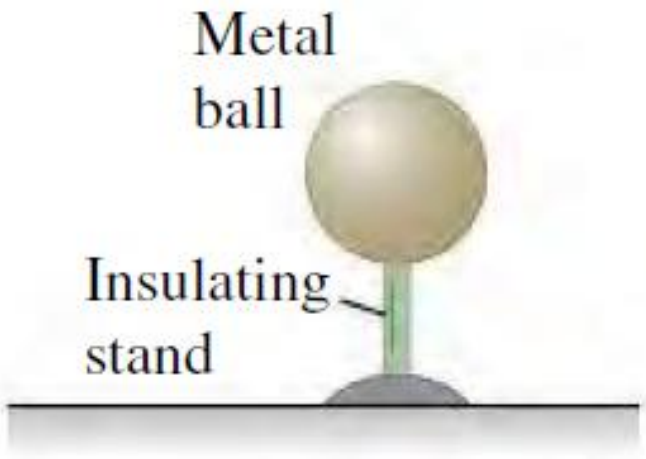
Grounding a Conductor

- The earth is a conductor, and it is so large that it can act as an infinite source of extra electrons or sink of unwanted electrons. The charge it acquires via induction will be equal and opposite to the charge remaining on the electrically induced body.

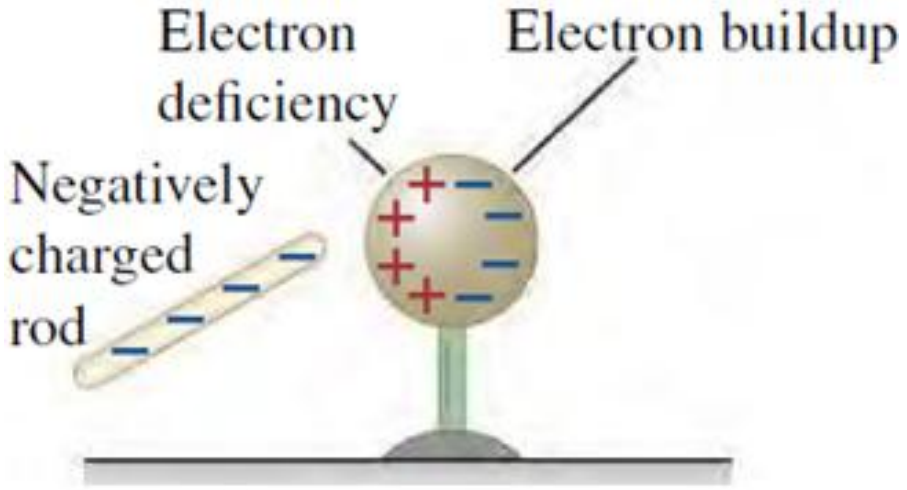
Charging a Conductor by Induction

- Charging by induction is the process in which a charged body can give another body a charge of opposite sign without losing any of its own charge.
- Excess charges that develop in the region of a body during electrical induction are called induced charges.
- In a metallic conductor, the mobile charges are always negative electrons.
- In ionic solutions and ionized gases, both positive and negative charges are mobile.

Charging a metal ball by induction

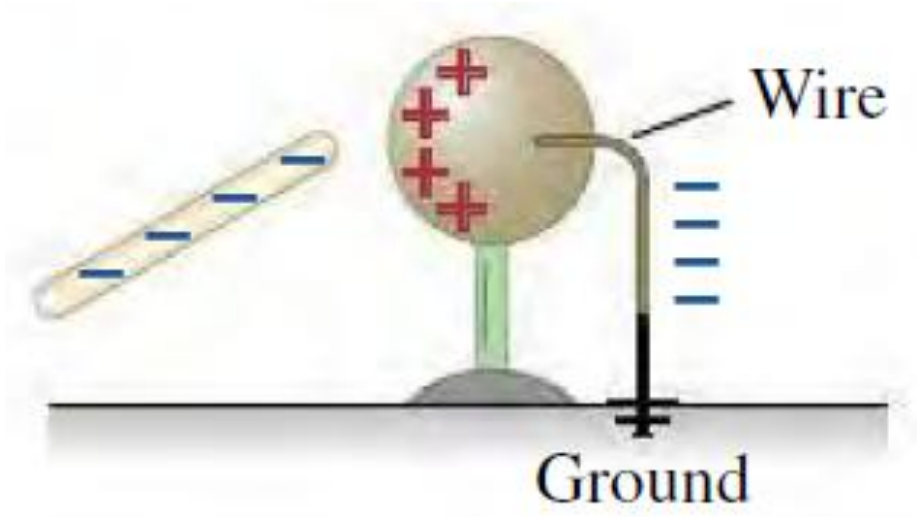


(a) Uncharged metal ball

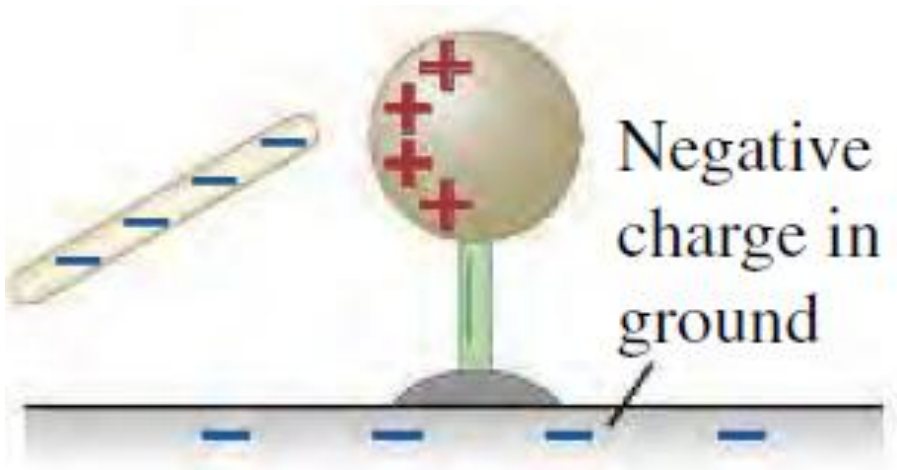


(b) Negative charge on rod repels electrons, creating zones of negative and positive induced charge.

Charging a metal ball by induction

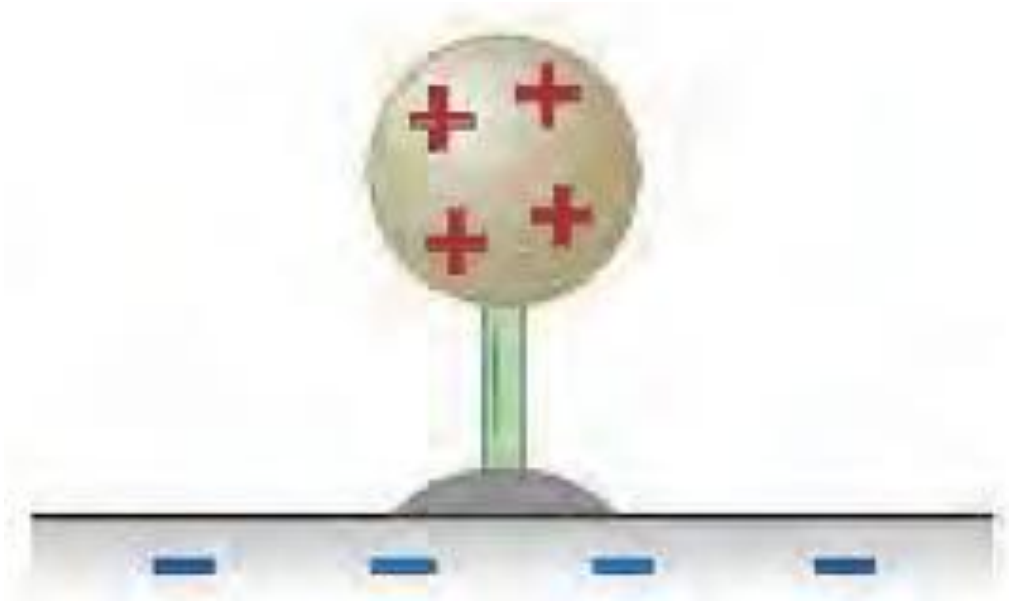


(c) Wire lets electron build-up (induced negative charge) flow into ground.



(d) Wire removed; ball now has only an electron-deficient region of positive charge.

Charging a metal ball by induction



(e) Rod removed; electrons rearrange themselves, ball has overall electron deficiency (net positive charge).

- Excess charges that develop in the region of a body during electrical induction are called **induced charges**.

- The earth is a conductor, and it is so large that it can act as an infinite source of extra electrons or sink of unwanted electrons. The charge it acquires via induction will be equal and opposite to the charge remaining on the electrically induced body.

- In a metallic conductor, the mobile charges are always negative electrons.

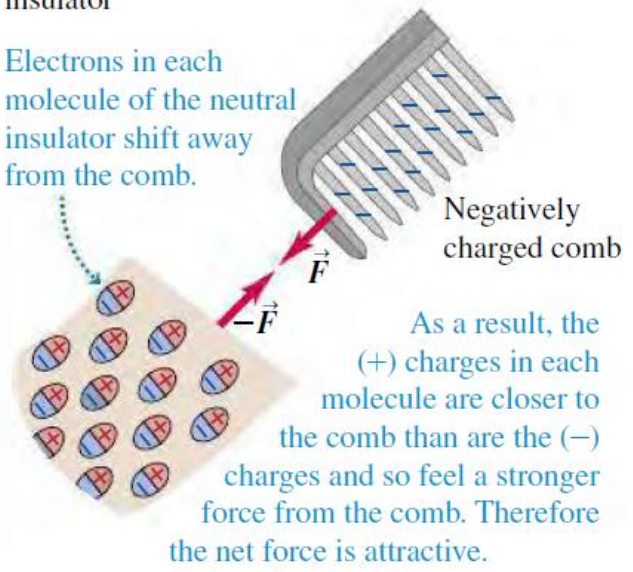
- In ionic solutions and ionized gases, both positive and negative charges are mobile.

- A charged body can exert forces even on objects that are *not* charged themselves – induced-charge effect.
- This is due to *polarization*, in which a charged object of *either* sign exerts an attractive force on an uncharged (neutral) insulator.
- An electrical neutral object may have regions of positive and negative charge within it, separated from one another. Such an object is polarized.
- A polarized object can experience an electric force even though its net charge is zero.

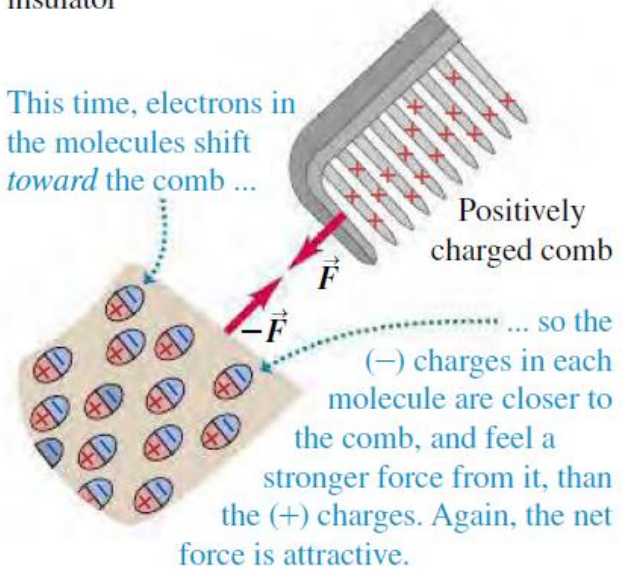
(a) A charged comb picking up uncharged pieces of plastic



(b) How a negatively charged comb attracts an insulator



(c) How a positively charged comb attracts an insulator

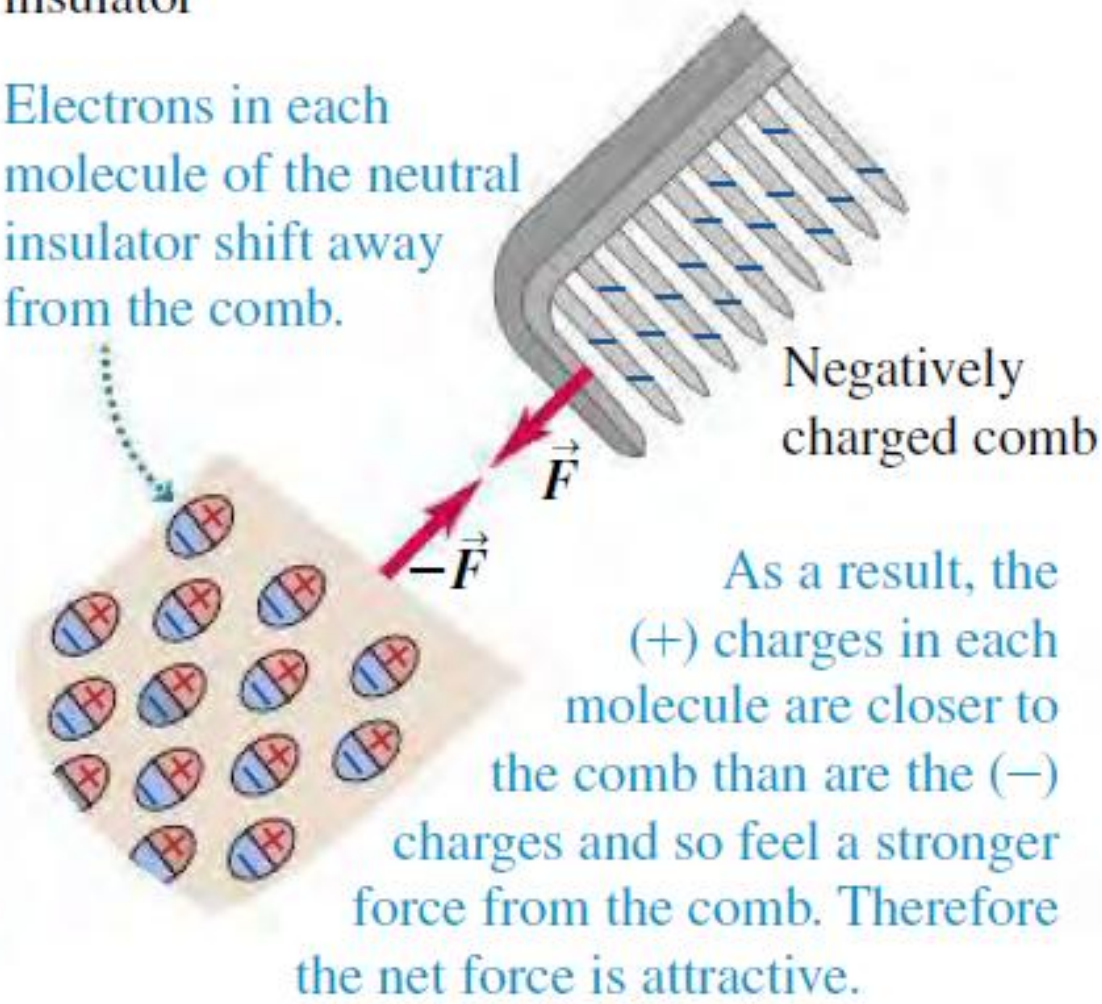


The charges within the molecules of an insulating material can shift slightly. As a result, a comb with either sign of charge attracts a neutral insulator. By Newton's third law the neutral insulator exerts an equal-magnitude attractive force on the comb.

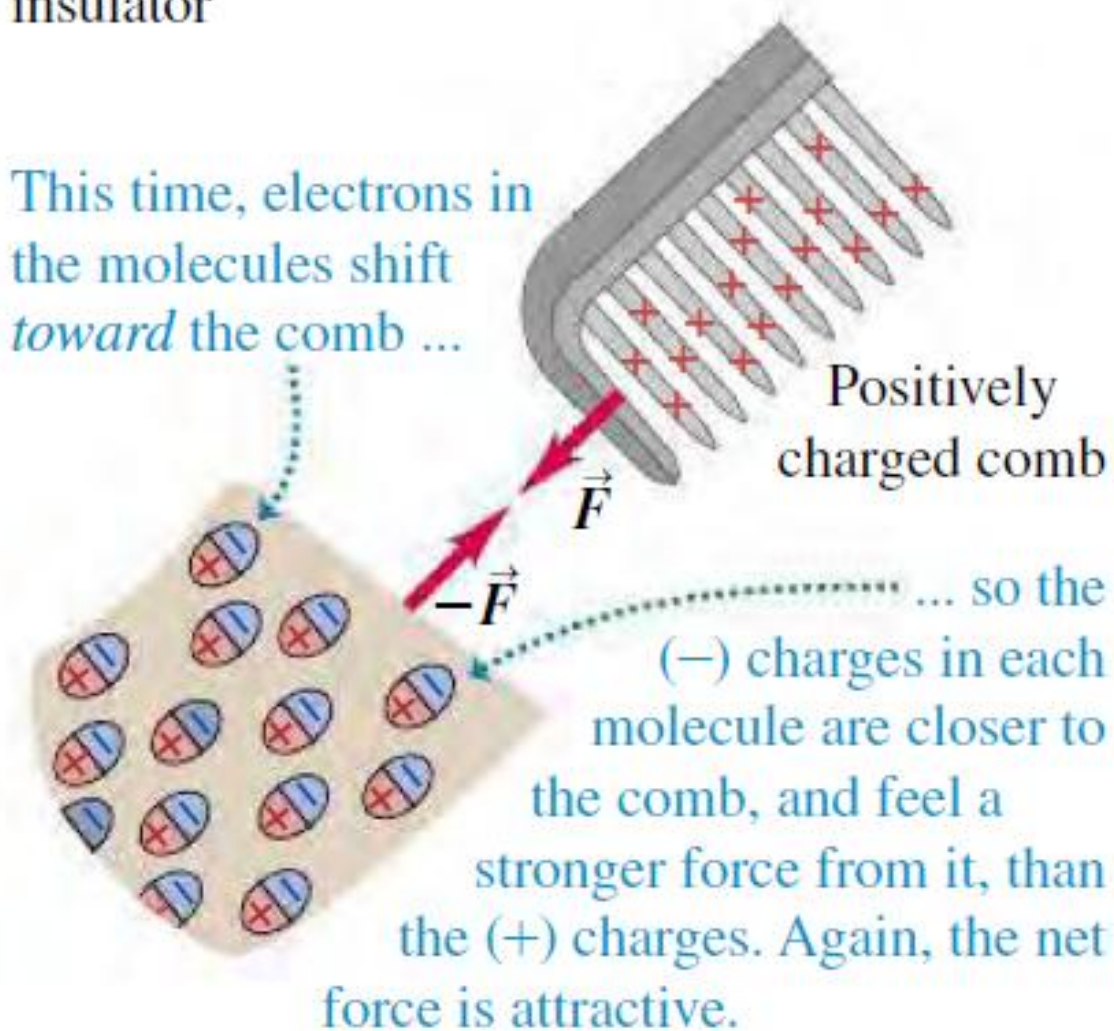
(a) A charged comb picking up uncharged pieces of plastic



(b) How a negatively charged comb attracts an insulator



(c) How a positively charged comb attracts an insulator



Chapter 21.3

Coulomb's Law

21.3 Coulomb's Law

Coulomb's Law states that:

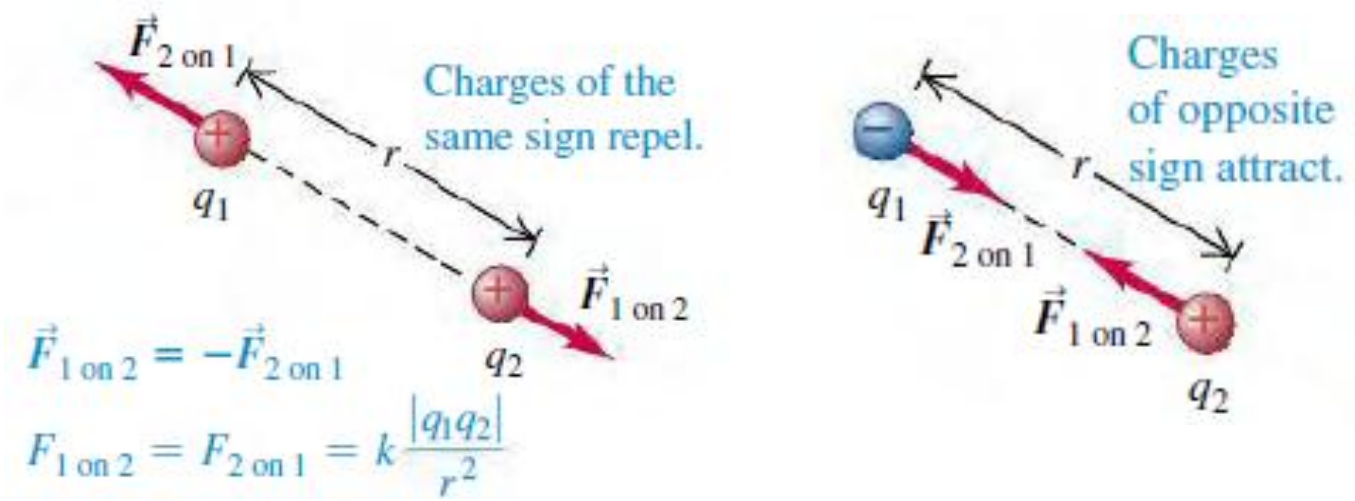
The magnitude of the electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.

$$F = k\frac{|q_1||q_2|}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{|q_1||q_2|}{r^2}$$

$$\left(k = \frac{1}{4\pi\varepsilon_0} = 9\times10^9\ N\cdot m/C^2\right)$$

21.3 Coulomb's Law

- The directions of the forces the two charges exert on each other are always along the line joining them, as shown in the figure.



Example 21.1 Electric force versus gravitational force

An α particle ("alpha") is the nucleus of a helium atom. It has a mass of $m = 6.66 \times 10^{-27}\ kg$ and a charge $q = +2e = 3.2 \times 10^{-19}\ C$.

Compare the force of the electric repulsion between two α particles with the force of gravitational attraction between them.

Identify and Set Up

The magnitude F_e of the electric force

$$F_e = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{r^2}$$

The magnitude F_g of the gravitational fore

$$F_g = G\frac{m^2}{r^2}$$

We compare these two magnitudes by calculating their ratio.

Execute

The ratio of the electric force to the gravitational force is

$$\frac{F_e}{F_g} = \frac{1}{4\pi\varepsilon_0 G}\frac{q^2}{m^2}$$
$$= \frac{9.0 \times 10^9\ N \cdot m^2/C^2}{6.67 \times 10^{-11}\ N \cdot m^2/kg^2}\frac{(3.2 \times 10^{-19}\ C)^2}{(6.64 \times 10^{-27}\ kg)^2}$$
$$= 3.1 \times 10^{35}$$

Evaluate

This astonishingly large number shows that the gravitational force in this situation is completely negligible in comparison to the electric force. This is always true for interactions of atomic and subatomic particles.

But within objects the size of a person or a planet, the positive and negative charges are nearly equal in magnitude, and the net electric force is usually much smaller than the gravitational force.

Problem-solving strategy 21.1: Coulomb's Law

IDENTIFY *the relevant concepts:* Coulomb's law describes the electric force between charged particles.

SET UP *the problem* using the following steps:

1. Sketch the locations of the charged particles and label each particle with its charge.

2. If the charges do not all lie on a single line, set up an xy coordinate system.

3. The problem will ask you to find the electric force on one or more particles. Identify which these are.

EXECUTE *the solution* as follows:

1. For each particle that exerts an electric force on a given particle of interest, find the magnitude of that force.

2. Using those magnitudes, sketch a free-body diagram showing the electric-force vectors acting on each particle of interest. The force exerted by particle 1 on particle 2 points from particle 2 toward particle 1 if the charges have opposite signs, but points from particle 2 directly away from particle 1 if the charge have the same sign.

EXECUTE *the solution* as follows:

3. Use the principle of superposition to calculate the total electric force——a vector sum——on each particle of interest.

4. Use consistent units; SI units are completely consistent. With $1/(4\pi\varepsilon_0) = 9.0 \times 10^9\ N \cdot m^2/C^2$, distances must be in meters, charges in coulombs, and forces in newtons.

EXECUTE *the solution* as follows:

5. Some examples and problems in this and later chapters involve continuous distributions of charge along a line, over a surface, or throughout a volume. In these cases the vector sum in step 3 becomes a vector integral. We divide the charge distribution into infinitesimal pieces, use Coulomb's law for each piece, and integrate to find the vector sum. Sometimes this can be done without actual integration.

EXECUTE *the solution* as follows:

6. Exploit any symmetries in the charge distribution to simplify your problem solving. For example, two identical charges q exert zero net electric force on a charge Q midway between them, because the forces on Q have equal magnitude and opposite direction.

EVALUATE your answer:

Check whether your numerical results are reasonable.

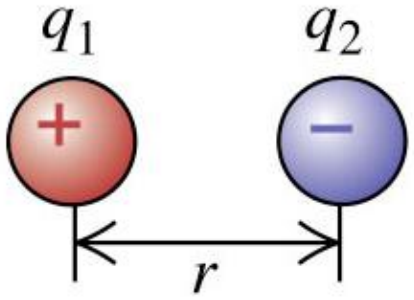
Confirm that the direction of the net electric force agrees with the principle that charges of the same sign repel and charges of opposite sign attract.

Example 21.2 Force between two point charges

Two point charges, $q_1 = +25\,nC$ and $q_2 = -75\,nC$, are separated by a distance of $3.0\,cm$ (Fig. 21. 10a). Find the magnitude and direction of

a) the electric force that q_1 exerts on q_2;

b) the electric force that q_2 exerts on q_1.



(a)

Identify and Set Up

We use Coulomb's law to calculate the magnitude of the force that each particle exerts on the other. The problem asks us for the force on each particle due to the other particle, so we use Newton's third law.

Execute

a) Converting charge to coulombs and distance to meters, the magnitude of the force that q_1 exerts on q_2 is

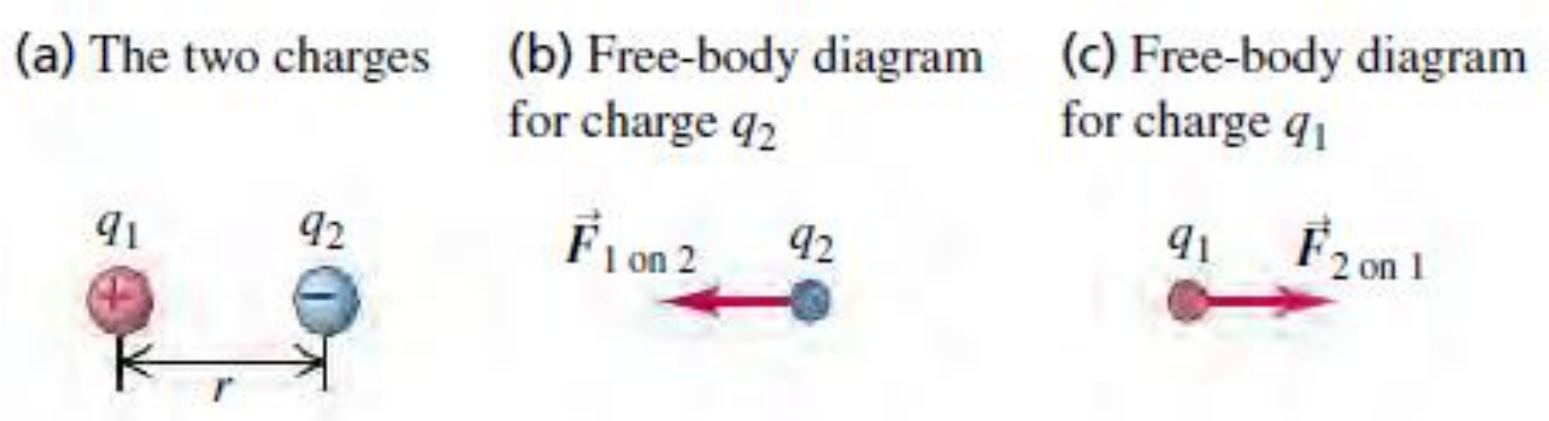
$$F_{1\ on\ 2} = \frac{1}{4\pi\varepsilon_0}\frac{|q_1 q_2|}{r^2}$$

$$= \left(9.0 \times 10^9\ N \cdot \frac{m^2}{C^2}\right)\frac{(+25 \times 10^{-9}\ C)(-75 \times 10^{-9}\ C)}{(0.030\ m)^2}$$

$$= -0.019\ N$$

Example 21.2 (SOLN)

Since the two charges have opposite signs, the force is attractive; that is, the force that acts on q_2 is directed toward q_1 along the line joining the two charges, as shown in the figure below.



Example 21.2 (SOLN)

b) Remember that Newton's third law applies to the electric force. Even though the charges have different magnitudes, the magnitude of the force that q_2 exerts on q_1 is the same as the magnitude of the force that q_1 exerts on q_2:

$$F_{2\ on\ 1} = 0.019\ N$$

Newton's third law also states that the direction of the force q_2 exerts on q_1 is exactly opposite the direction of the force that q_1 exerts on q_2.

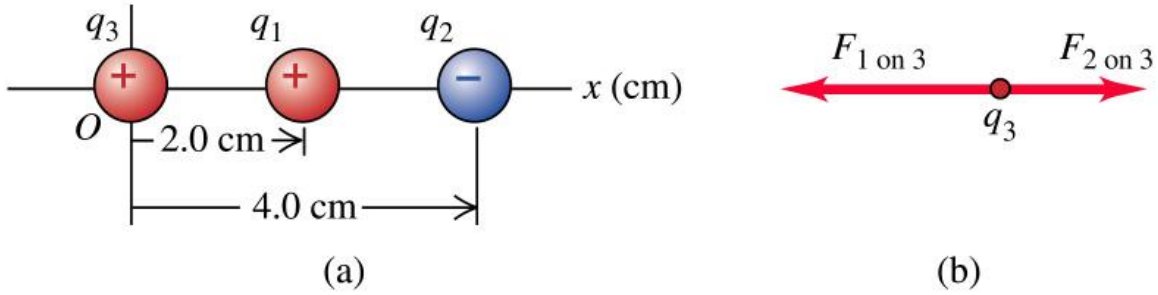
Evaluate

Note that the force on q_1 is directed toward q_2, as it must be, since charges of opposite sign attract each other.

Example 21.3 Vector addition of electric forces on a line

Two point charges are located on the positive x-axis of a coordinate system (Fig. 21.11a). Charge $q_1 = 1.0\,nC$ is $2.0\,cm$ from the origin, and charge $q_2 = -3.0\,nC$ is $4.0\,cm$ from the origin.

What is the total force exerted by these two charges on a charge $q_3 = 5.0\,nC$located at the origin? Gravitational forces are negligible.



Identify

Here there are two electric forces acting on the charge q_3, and we must add these forces to find the total force.

Set Up

Figure 21.11a shows the coordinate system. Our target variable is the net electric force exerted on charge q_3 by the other two charges. This is the vector sum if the forces due to q_1 and q_2 individually.

Execute

Figure 21.11b is a free-body diagram for charge q_3. Note that q_3 is repelled by q_1 (which has the same sign) and attracted to q_2 (which has the opposite sign).

Converting charge to coulombs and distance to meters, we use Coulomb's Law to find the magnitude $F_{1\ on\ 3}$ of the force of q_1 on q_3:

Execute

$$F_{1\ on\ 3} = \frac{1}{4\pi\varepsilon_0}\frac{|q_1 q_2|}{r^2}$$
$$= \left(9.0 \times 10^9\ N \cdot \frac{m^2}{C^2}\right)\frac{(1.0 \times 10^{-9}\ C)(5.0 \times 10^{-9}\ C)}{(0.020\ m)^2}$$
$$= 1.12 \times 10^{-4}\ N = 112\ \mu N$$

This force has a negative x-component because q_3 is repelled (that is, pushed in the negative x-direction) by q_1

Execute

The magnitude $F_{2\ on\ 3}$ of the force of q_2 on q_3 is

$$F_{2\ on\ 3} = \frac{1}{4\pi\varepsilon_0}\frac{|q_2 q_3|}{r^2}$$

$$= \left(9.0 \times 10^9\ N \cdot \frac{m^2}{C^2}\right)\frac{(3.0 \times 10^{-9}\ C)(5.0 \times 10^{-9}\ C)}{(0.020\ m)^2}$$

$$= 8.4 \times 10^{-5}\ N = 84\ \mu N$$

This force has a positive x-component because q_3 is attracted (that is ,pulled in the positive x-direction) by q_2.

Execute

The sum of the x-components is

$$F_x = -112\ \mu N + 84\ \mu N = -28\ \mu N$$

There are no $y-$ or $z-$components. Thus the total force on q_3 is directed to the left, with magnitude $28\ \mu N = 2.8 \times 10^{-5}\ N$

Evaluate

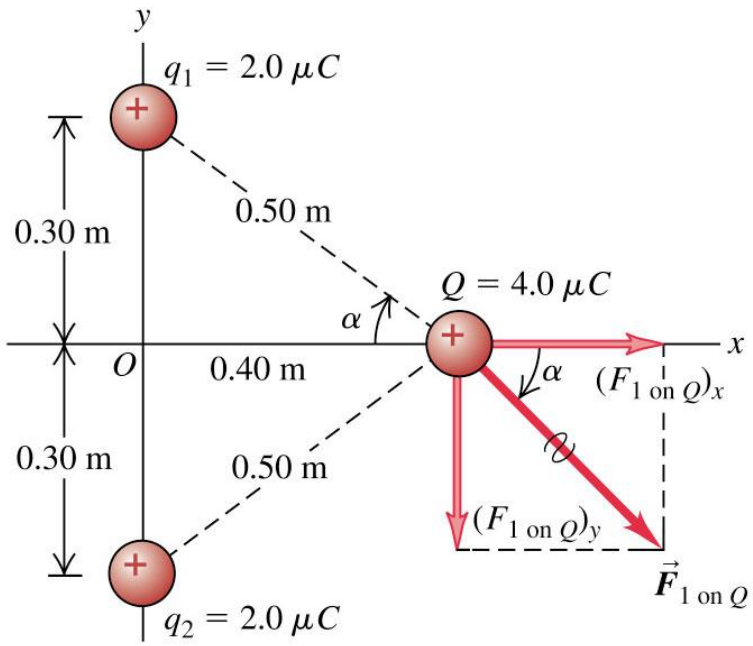
To check the magnitudes of the individual forces, note that q_2 has three times as much charge (in magnitude) as q_1 but is twice as far from q_3.

From Eq. (21.2), this means that $F_{2\,on\,3}$ must be $3/2^2 = 3/4$ as large as $F_{1\,on\,3}$. Indeed, our results show that this ratio is $(84\,\mu N)/(112\,\mu N) = 0.75$.

The direction of the net force also makes sense: $\vec{F}_{1\,on\,3}$ is opposite to and has a larger magnitude than $\vec{F}_{2\,on\,3}$, so the net force is in the direction of $\vec{F}_{1\,on\,3}$.

Example 21.4 Vector addition of electric forces in a plane

In the figure below, two equal positive charges $q_1 = q_2 = 2.0\,\mu C$ interact with a third point charge $Q = 4.0\,\mu C$. Find the magnitude and direction of the total (net) force on Q.



Identify and Set up

As in Example 21.3, we have to compute the force that each charge exerts on Q and then find the vector sum of the forces. The best way to do this is to use components.

Execute

The figure above shows the force on Q due to the upper charge q_1. From Coulomb's law, the magnitude F of this force is

$$F_{1\ on\ Q} = \left(9.0 \times 10^9\ N \cdot \frac{m^2}{C^2}\right) \frac{(4.0 \times 10^{-6} C)(2.0 \times 10^{-6} C)}{0.50\ m^2}$$

$$= 0.29\ N$$

Execute

The angle α is below the x-axis, so the components of this force are given by

$$(F_{1\ on\ Q})_x = \left(F_{1\ on\ Q}\right)\cos\alpha = (0.29\ N)\frac{0.40\ m}{0.50\ m} = 0.23\ N$$

$$(F_{1\ on\ Q})_y = \left(F_{1\ on\ Q}\right)\sin\alpha = -(0.29\ N)\frac{0.30\ m}{0.50\ m} = -0.17\ N$$

The lower charge q_2 exerts a force with the same magnitude but at an angle α above the x-axis.

Execute

From symmetry we see that its x-component is the same as that due to the upper charge, but its y-component has the opposite sign. So the components of the total force $\vec{F}$ on Q are

$$F_x = 0.23\ N + 0.23\ N = 0.46\ N$$

$$F_y = -0.17\ N + 0.17\ N = 0$$

The total force on Q is in the $+x$-direction, with magnitude $0.46\ N$

Evaluate

The total force on Q is in a direction that points neither directly away from q_1 nor directly away from q_2.

Rather, this direction is a compromise that points away from the system of charges q_1 and q_2.

Can you see that the total force would not be in the $+x$-direction if q_1 and q_2 were not equal or if the geometrical arrangement of the changes were not so symmetrical?

Chapter 21.4
Electric Field and Electric Forces

21.4 Electric Field and Electric Forces

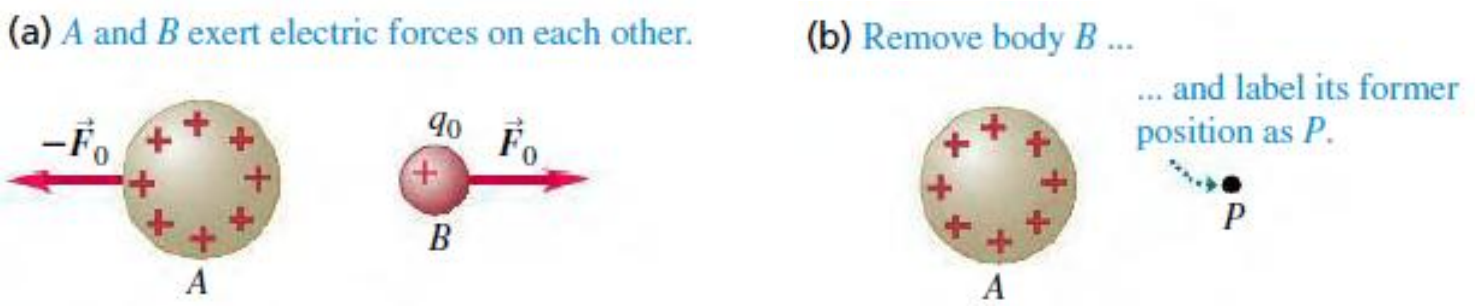
Electric Field $\vec{E}$

- The electric field at a point is defined as the electric force $\vec{F}_0$ experienced by a test charge q_0 at the point, divided by the charge q_0. (The electric field at a certain point is equal to the electric force per unit charge experienced by a charge at that point)

$$\vec{E} = \frac{\vec{F}_0}{q_0} \qquad (21.3)$$

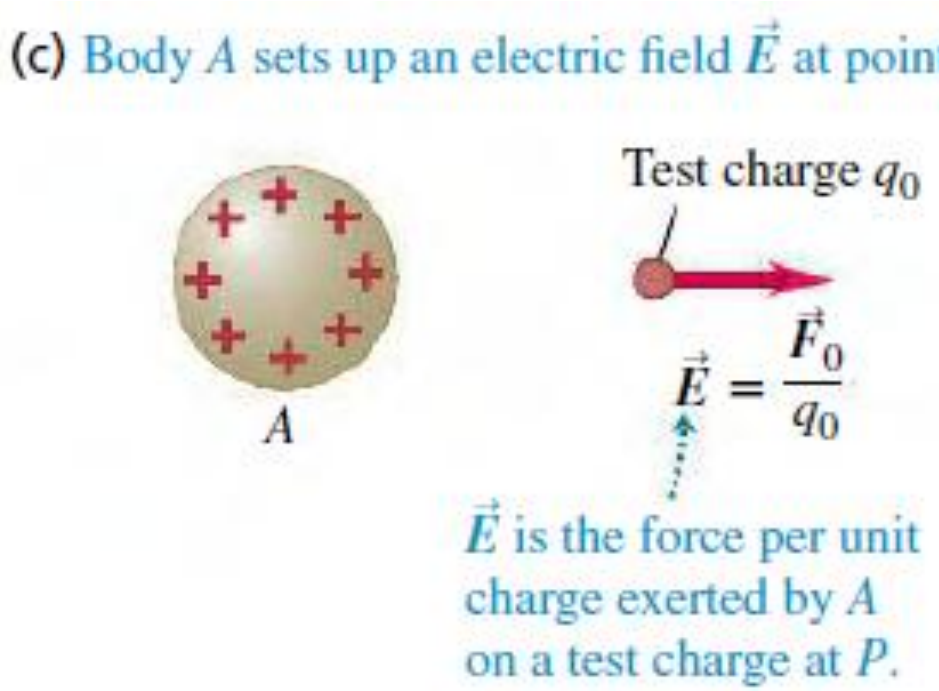
21.4 Electric Field and Electric Forces

- SI unit is 1 newton per coulomb ($1\ N/C$).
- The figure shows how a charged body creates an electric field in the space around it.



21.4 Electric Field and Electric Forces

- The figure shows how a charged body creates an electric field in the space around it.



Characteristics of electric field

- The electric field that a charged body produces exists at all points in the region around itself.
- The electric force is an "action-at-a-distance" force that acts across empty space without needing any matter to transmit it through the intervening space.
- The electric force on a charged body is exerted by the electric field created by other charged bodies.

21.4 Electric Field and Electric Forces

- The electric field $\vec{E}$, or electric force per unit charge, is useful because it does not depend on the charge of the body on which the electric force is exerted.

Chapter 21 Electric Charge and Electric Field

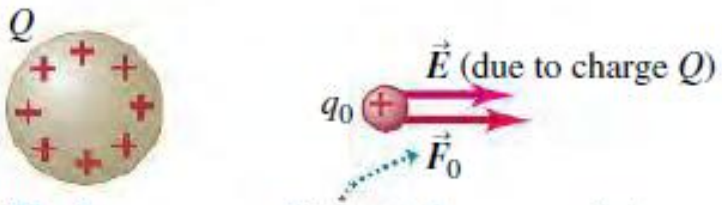
- The electric force experienced by a test charge q_0 can vary from point to point, so the electric field can also be different at different points.

- For this reason, Eq. (21.4) can be used only to find the electric force on a point charge, not on large bodies where the electric field may vary with different locations on the body.

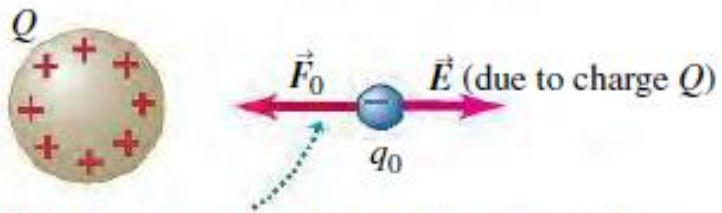
21.4 Electric Field and Electric Forces

The figure shows the electric force exerted by an electric field on a point charge.

$$\vec{F}_0 = q_0\vec{E} \qquad (21.4)$$



The force on a positive test charge q_0 points in the direction of the electric field.



The force on a negative test charge q_0 points opposite to the electric field.

Electric Field of a Point Charge

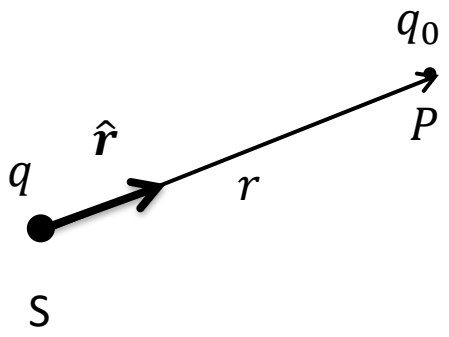
- The magnitude electric field of a point charge q:

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \qquad (21.6)$$

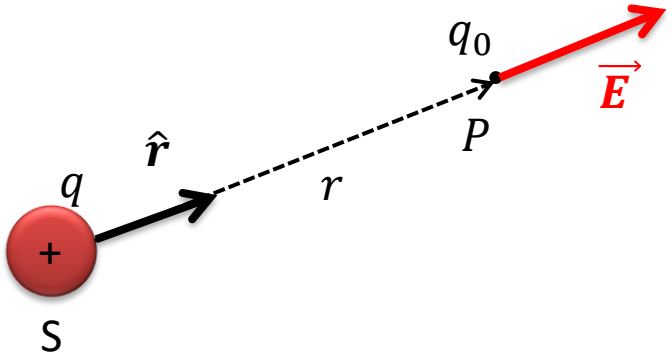
- The source point S is the location of the point charge that is producing the electric field we are investigating.
- The field point P is the location where we are determining the field.
- The unit vector $\hat{\boldsymbol{r}}$ denotes the direction along the line from source point to field point.

Electric Field of a Point Charge

The electric field $\vec{E}$ produced at point P by an isolated point charge q at S.



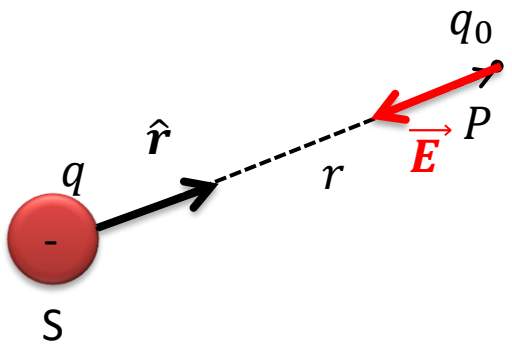
Unit vector $\hat{\boldsymbol{r}}$ points from source point S to field point P.



At each point P, the electric field set up by an isolated positive point charge q points directly away from the charge in the direction as $\hat{\boldsymbol{r}}$.

Electric Field of a Point Charge

The electric field $\vec{E}$ produced at point P by an isolated point charge q at S.



At each point P, the electric field set up by an isolated negative point charge q points directly toward the charge in the opposite direction as $\hat{\boldsymbol{r}}$.

21.4 Electric Field and Electric Forces

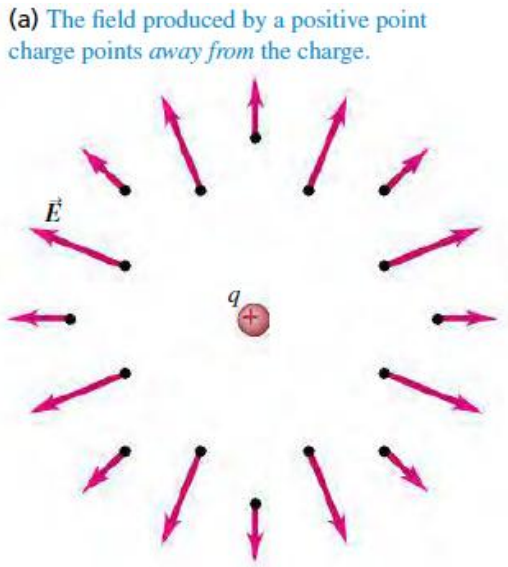
- The magnitude and direction of the electric field $\vec{E}$ can be given by the vector equation:

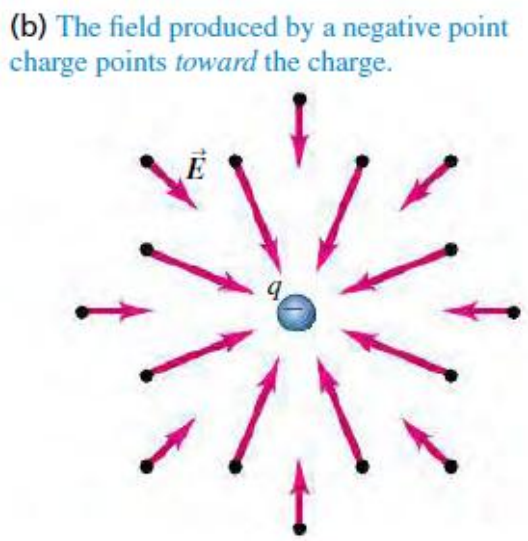
$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} \qquad (21.7)$$

- The **vector field** is thus the infinite set of vectors associated with every point in space in an electric field.

21.4 Electric Field and Electric Forces

The following figure shows that a point charge q produces an electric field $\vec{E}$ at all points in space. The field strength decreases with increasing distance.





21.4 Electric Field and Electric Forces

- When the magnitude and direction of the field (and hence its vector components) are the same everywhere throughout an electric field, the field is uniform.

Example 21.5 Electric-field magnitude for a point charge

What is the magnitude of the electric field at a field point $2.0\,m$ from a point charge $q = 4.0\,nC$? (The point charge could represent any small charged object with this value of q, provided the dimensions of the object are much less than the distance from the object to the field point.)

Identify and Set Up

We are given the magnitude of the charge and the distance from the object to the field point, so we use Eq. (21.6) to calculate the field magnitude E.

Execute

From Eq. (21.6),

$$E = \frac{1}{4\pi\varepsilon_0}\frac{|q|}{r^2} = \left(9.0\times10^9\ N\cdot\frac{m^2}{C^2}\right)\frac{4.0\times10^{-9}\ C}{(2.0\ m)^2}$$
$$= 9.0N/C$$

Evaluate

To check our result, we use the definition of electric field as the electric force per unit charge.

We can first use Coulomb's law to find the magnitude F_0 of the force on a test charge q_0 placed $2.0\,m$ from q:

$$F_0 = \frac{1}{4\pi\varepsilon_0}\frac{|qq_0|}{r^2}$$
$$= \left(9.0\times10^9\ N\cdot\frac{m^2}{C^2}\right)\frac{4.0\times10^{-9}\ C|q_0|}{(2.0\,m)^2}$$
$$= (9.0\ N/C)|q_0|$$

Evaluate

Then, from Eq. (21.3), the magnitude of $\vec{E}$ is:

$$E = \frac{F_0}{|q_0|} = 9.0\ N/C$$

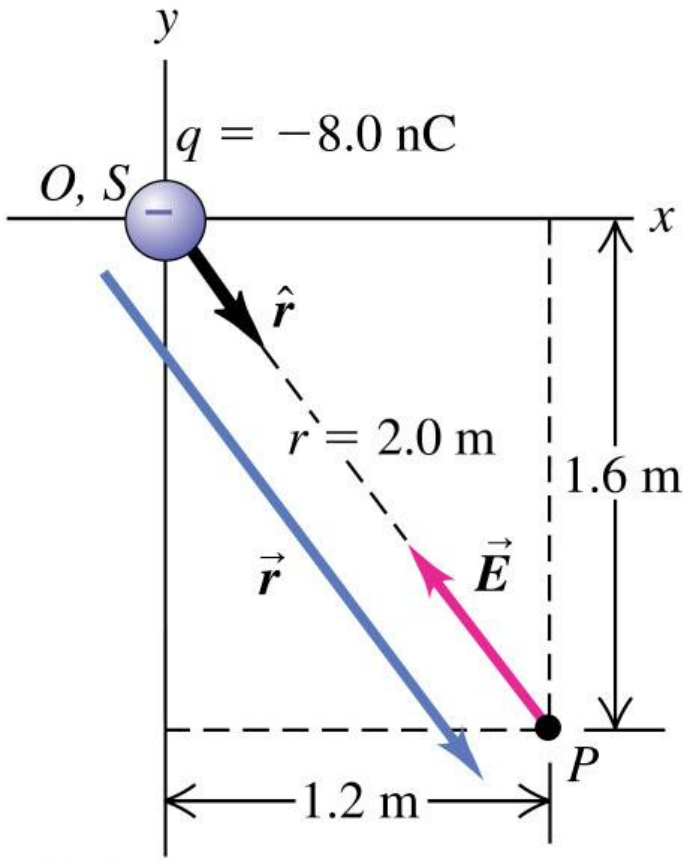
Because q is positive, the direction of $\vec{E}$ at this point is along the line from q towards q_0, as shown in Fig. 21. 17b.

However, the magnitude and direction of $\vec{E}$ do not depend on the sign of q_0.

Example 21.6 Electric-field vector for a point charge

A point charge $q = -8.0\,nC$ is located at the origin.

Find the electric-field vector at the field point $x = 1.2\,m, y = -1.6\,m.$



Identify and Set Up

The electric field is given in vector form by Eq. (21.7).

To use this equation, we utilize the coordinate system shown in the figure above to find the distance from the source point S (the position of the charge q) to the field point P, as well as the unit vector $\hat{\boldsymbol{r}}$ that points in the direction from S to P.

Execute

The distance from the charge at the source point S (which in this example is at the origin O) to the field point P is

$$r = \sqrt{x^2 + y^2} = \sqrt{(1.2\ m)^2 + (-1.6\ m)^2} = 2.0\ m$$

The unit vector $\hat{\boldsymbol{r}}$ is directed from the source point to the field point.

$$\hat{\boldsymbol{r}} = \frac{\vec{\boldsymbol{r}}}{r} = \frac{x\,\hat{\boldsymbol{\imath}} + y\,\hat{\boldsymbol{\jmath}}}{r}$$

$$= \frac{(1.2\ m)\,\hat{\boldsymbol{\imath}} + (-1.6\ m)\,\hat{\boldsymbol{\jmath}}}{2.0\ m} = 0.60\,\hat{\boldsymbol{\imath}} - 0.80\,\hat{\boldsymbol{\jmath}}$$

Execute

Hence the electric field vector is:

$$\vec{\boldsymbol{E}} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$

$$= (9.0 \times 10^9\ N \cdot m^2/C^2\ )$$

$$\frac{(-8.0 \times 10^{-9}\ C)}{(2.0\ m)^2}(0.60\ \hat{\boldsymbol{\imath}} - 0.80\ \hat{\boldsymbol{\jmath}})$$

$$= (-11\ N/C)\hat{\boldsymbol{\imath}} + (14\ N/C\ \hat{\boldsymbol{\jmath}})$$

Evaluate

Since q is negative, $\vec{E}$ points from the field point to the charge (the source point), in the direction opposite to $\hat{r}$.

Example 21.7 Electron in a uniform field

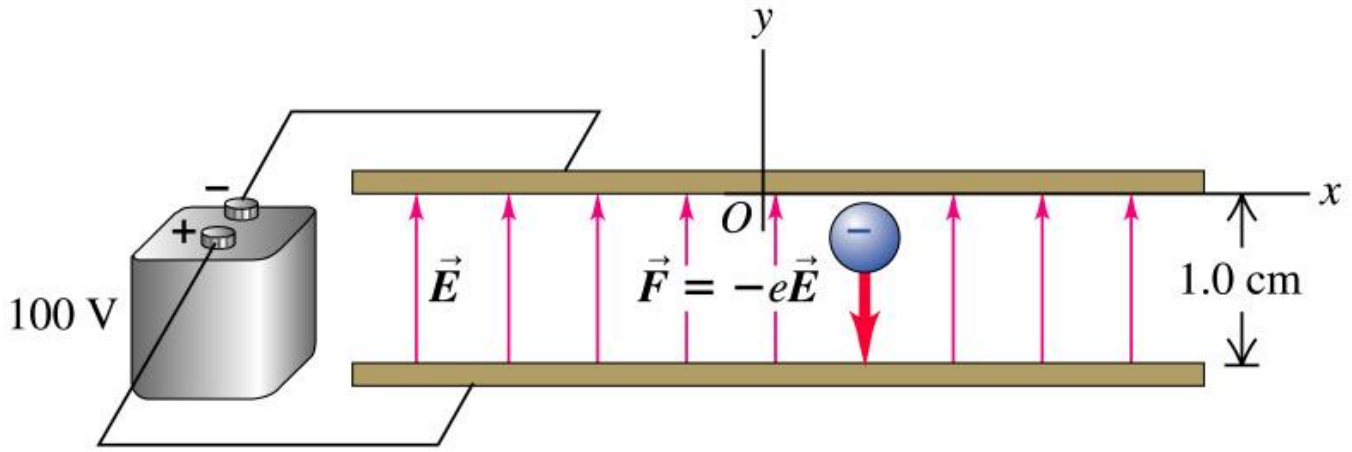
When the terminals of a battery are connected to two large parallel conducting plates, the resulting charges on the plates cause an electric field $\vec{E}$ in the region between the plates that is very nearly uniform. (We will see the reason for this uniformity in the next section. Charged plates of this kind are used in common electrical devices called capacitors, to be discussed in Chapter 24.) If the plates are horizontal and separated by $1.0\ cm$ and the plates are connected to a 100-volt battery, the magnitude of the field is $E = 1.00 \times 10^4\ N/C$. Suppose the direction of $\vec{E}$ is vertically upward, as shown by the vectors in the following figure.

Example 21.7 Electron in a uniform field

a) If an electron is released from rest at the upper plate, what is its acceleration?
b) What speed and kinetic energy does it acquire while traveling $1.0\ cm$ to the lower plate?
c) How much time is required for it to travel this distance? An electron has a charge $-e = -1.60 \times 10^{-19} C$ and a mass $m = 9.11 \times 10^{-31}\ kg$.

Identify

This example involves several concepts: the relation between electric field and electric force, the relation between force and acceleration, the definition of kinetic energy, and the kinematic relations between acceleration, distance, velocity, and time.



Set Up

The figure above shows a coordinate system.

We are given the electric field, so we use Eq. (21.4) to find the force on the electron and Newton's second law to find its acceleration.

Because the field is uniform between the plates, the force and acceleration are constant and we can use the constant-acceleration formulas from Chapter 3 to find the electron's velocity and travel time. We find the kinetic energy using the definition $K = \frac{1}{2}mv^2$.

Execute

a) Note that $\vec{E}$ is upward (in the $+y$-direction) but $\vec{F}$ is downward because the charge of the electron is negative. Thus F_y is negative.

Because F_y is constant, the electron moves with constant acceleration a_y given by:

$$a_y = \frac{F_y}{m} = \frac{-eE}{m}$$

$$= \frac{(-1.60 \times 10^{-19}\ C)(1.00 \times 10^{4}\ N/C)}{9.11 \times 10^{-31}\ kg} = -1.76 \times 10^{15}\ m/s^2$$

Execute

This is an enormous acceleration! To give a $1000-kg$ car this acceleration, we would need a force of about $2\times10^{18}\,N$ (about $2\times10^{14}\,tons$).

The gravitational force on the electron is completely negligible compared to the electric force.

Execute

b) The electron starts from rest, so its motion is in the y-direction only (the direction of the acceleration).

We can find the electron's speed at any position using the constant-acceleration formula $v_y{}^2 = v_{0y}{}^2 + 2a_y(y - y_0)$.

We have $v_{0y} = 0$ and $y_0 = 0$, so the speed $|v_y|$ when $y = -1.0\,cm = -1.0 \times 10^{-2}\,m$ is:

Execute

$$|v_y| = \sqrt{2a_y y}$$
$$= \sqrt{2(-1.76 \times 10^{15}\ m/s^2)(-1.0 \times 10^{-2}\ m)}$$
$$= 5.9 \times 10^6\ m/s$$

The velocity is downward, so its y-component is $v_y = -5.9 \times 10^6\ m/s$. The electron's kinetic energy is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}(9.11 \times 10^{-31}\ kg)(5.9 \times 10^6\ m/s)^2$$
$$= 1.6 \times 10^{-17}\ J$$

Execute

c) From the constant acceleration formula $v_y = v_{0y} + a_y t$, we find that the time required is very brief:

$$t = \frac{v_y - v_{0y}}{a_y}$$
$$= \frac{(-5.9 \times 10^6\ m/s) - (0\ m/s)}{-1.76 \times 10^{15}\ m/s^2}$$
$$= 3.4 \times 10^{-9}\ s$$

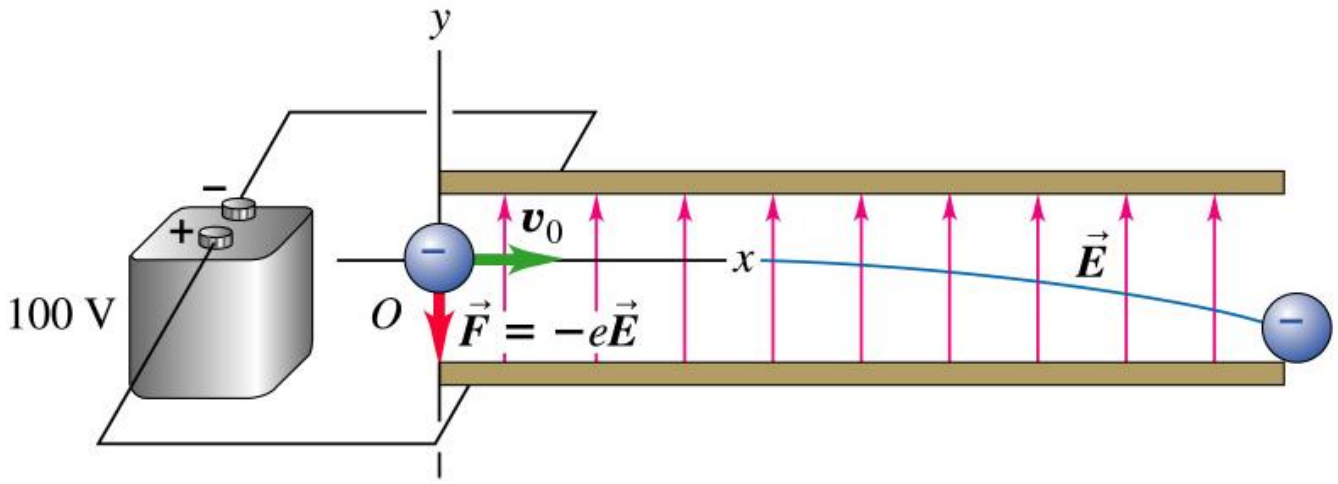
(we could also have found the time by solving the equation $y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$ for t.)

Evaluate

This example shows that when working problems about subatomic particles such as electrons, many quantities – including acceleration, speed, kinetic energy, and time – will have very different values from what we have seen for ordinary objects such as baseballs and automobiles.

Example 21.8 An electron trajectory

If we launch an electron into the electric field of Example 21.7 with an initial horizontal velocity v_0 , what is the equation of its trajectory?



Identify and Set Up

The force and acceleration are constant, which is the same as in Example 21.7, and there is no acceleration in the x-direction.

Hence we can use the kinematic equations from Chapter 3 for two-dimensional motion with constant acceleration.

Execute

We have $a_x = 0$ and $a_y = (-e)E/m$.

At $t = 0$, $x_0 = y_0 = 0$, $v_{0x} = v_0$, and $v_{0y} = 0$.

Hence at time t,

$$x = v_0 t, y = \frac{1}{2} a_y t^2 = -\frac{1}{2}\frac{eE}{m} t^2$$

Eliminating t between these equations, we get

$$y = -\frac{1}{2}\frac{eE}{m{v_0}^2} x^2$$

Evaluate

This is the equation of a parabola, just like the trajectory of a projectile launched horizontally in the earth's gravitational field.

For a given initial velocity of the electron, the curvature of the trajectory depends on the field magnitude E.

If we reverse the signs of the charges on the two plates in Fig. 21.21, the direction of $\vec{E}$ reverses, and the electron trajectory will curve up, not down.

Evaluate

Hence we can "steer" the electron by varying the charges on the plates.

The electric field between charged conducting plates is used in this way to control the trajectory of electron beams in oscilloscopes.

Chapter 21.5

Electric-Field Calculations

21.5 Electric-Field Calculations

Principle of Superposition of Electric Fields

- In most realistic situations that involve electric fields and forces, charge is distributed over space.
  - λ, linear charge density, is used when charge is distributed along a line and has units of C/m.
  - σ, surface charge density, refers to charge that is distributed over a surface, with units C/m^2.
  - ρ, volume charge density, represents charge when it is distributed throughout a volume and its units are C/m^3.

21.5 Electric-Field Calculations

Principle of Superposition of Electric Fields

- Thus the principle of superposition of electric fields is used to find the total electric field at a given point P, set up by the individual electric fields produced by each point charge in the field:

$$\vec{\boldsymbol{E}} = \frac{\vec{\boldsymbol{F}}_0}{q_0} = \vec{\boldsymbol{E}}_1 + \vec{\boldsymbol{E}}_2 + \vec{\boldsymbol{E}}_3 + \cdots$$

Example 21.9 Field of an electric dipole

Point charges q_1 and q_2 of $+12\,nC$ and $-12\,nC$, respectively, are placed $0.10\,m$ apart (Fig. 21.23). This combination of two charges with equal magnitude and opposite sign is called an electric dipole. (Such combinations occur frequently in nature. We'll study dipoles in more detail in Section 21.7.)

Compute the electric field caused by q_1, the field caused by q_2, and the total field a) at point a; b) at point b; c) at point c.

Chapter 21 Ele

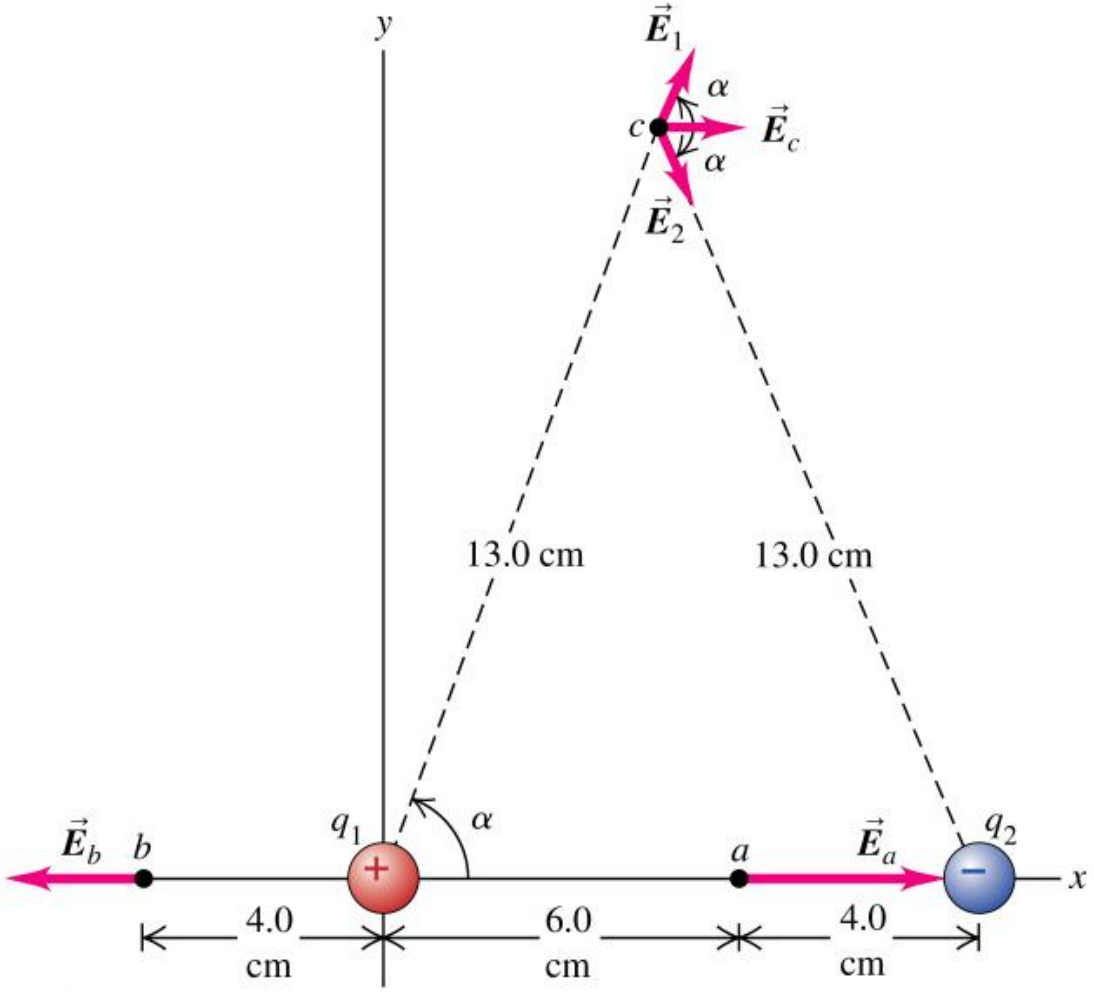


Fig. 21.23

Identify

We need to find the total electric field at three different points due to two point charges. We will use the principle of superposition: $\vec{\boldsymbol{E}} = \vec{\boldsymbol{E}}_1 + \vec{\boldsymbol{E}}_2$.

Set up

Figure 21.23 shows the coordinate system and the locations of the three field points, a, b, and c.

Execute

a) At point a the field $\vec{E}_1$ caused by the positive charge q_1 and the field $\vec{E}_2$ caused by the negative charge q_2 are both directed toward the right. The magnitudes of $\vec{E}_1$ and $\vec{E}_2$ are:

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{|q_1|}{r^2} = (9.0\times10^9\ N\cdot m^2/C^2)\frac{12\times10^{-9}\ C}{(0.060\ m)^2} = 3.0\times10^4\ N/C$$

$$E_2 = \frac{1}{4\pi\varepsilon_0}\frac{|q_2|}{r^2} = (9.0\times10^9\ N\cdot m^2/C^2)\frac{12\times10^{-9}\ C}{(0.040\ m)^2} = 6.8\times10^4\ N/C$$

Execute

The components of $\vec{E}_1$ and $\vec{E}_2$ are

$$E_{1x} = 3.0 \times 10^4\ N/C, E_{1y} = 0$$

$$E_{2x} = 6.8 \times 10^4\ N/C, E_{2y} = 0$$

Hence at point a, the total electric field $\vec{E}_a = \vec{E}_1 + \vec{E}_2$ has components

$$(E_a)_x = E_{1x} + E_{2x} = (3.0 + 6.8) \times 10^4 N/C$$

$$(E_a)_y = E_{1y} + E_{2y} = 0$$

Execute

At point a the total field has magnitude $9.8 \times 10^4\ N/C$ and is directed toward the right, so

$$\vec{\boldsymbol{E}}_a = (9.8 \times 10^4\ N/C)\,\hat{\boldsymbol{\imath}}$$

b) At point b, the field $\vec{\boldsymbol{E}}_1$ due to q_1 is directed toward the left, while the field $\vec{\boldsymbol{E}}_2$ due to q_2 is directed toward the right. The magnitudes of $\vec{\boldsymbol{E}}_1$ and $\vec{\boldsymbol{E}}_2$ are:

Execute

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{|q_1|}{r^2} = (9.0\times10^9\ N\cdot m^2/C^2)\frac{12\times10^{-9}\ C}{(0.040\ m)^2} = 6.8\times10^4\ N/C$$

$$E_2 = \frac{1}{4\pi\varepsilon_0}\frac{|q_2|}{r^2} = (9.0\times10^9\ N\cdot m^2/C^2)\frac{12\times10^{-9}\ C}{(0.140\ m)^2} = 0.55\times10^4\ N/C$$

The components of $\vec{E}_1$, $\vec{E}_2$ and the total field $\vec{E}_b$ at point b are

$$E_{1x} = -6.8\times10^4\ N/C, \qquad E_{1y} = 0$$

Execute

$$E_{2x} = 0.55 \times 10^4\ N/C, \qquad E_{2y} = 0$$

$$(E_b)_x = E_{1x} + E_{2x} = (-6.8 + 0.55) \times 10^4\ N/C$$

$$(E_b)_y = E_{1y} + E_{2y} = 0$$

That is, the electric field at b has magnitude $6.2 \times 10^4\ N/C$ and is directed toward the left, so

$$\vec{\boldsymbol{E}}_b = (-6.2 \times 10^4\ N/C)\ \hat{\boldsymbol{\imath}}$$

Execute

c) At point c, both $\vec{E}_1$ and $\vec{E}_2$ have the same magnitude, since this point is equidistant from both charges and the charge magnitudes are the same:

$$E_1 = E_2 = \frac{1}{4\pi\varepsilon_0}\frac{|q|}{r^2}$$

$$= (9.0 \times 10^9\ N \cdot m^2/C^2)\frac{12 \times 10^{-9}\ C}{(0.13\ m)^2}$$

$$= 6.39 \times 10^4\ N/C$$

Execute

The directions of $\vec{E}_1$ and $\vec{E}_2$ are shown in the figure of this problem. The x-components of both vectors are the same:

$$E_{1x} = E_{2x} = E_1 \cos\alpha$$
$$= (6.39 \times 10^3\ N/C)\left(\frac{5}{13}\right) = 2.46 \times 10^3\ N/C$$

From symmetry the y-component E_{1y} and E_{2y} are equal and opposite and so add to zero. Hence the components of the total field $\vec{E}_c$ are:

Execute

$$(E_c)_x = E_{1x} + E_{2x} = 2(2.46 \times 10^4\ N/C) = 4.9 \times 10^3\ N/C$$

$$(E_c)_y = E_{1y} + E_{2y} = 0$$

So at point c the total electric field has magnitude $4.9 \times 10^3\ N/C$ and is directed toward the right, so

$$\vec{\boldsymbol{E}}_c = (4.9 \times 10^3\ N/C)\,\hat{\boldsymbol{\imath}}$$

Does it surprise you that the field at point c is parallel to the line between the two charges?

Evaluate

An alternative way to find the electric field at c is to use the vector expression for the field of a point charge, Eq. (21.7). The displacement vector $\vec{r}_1$ from q_1 to point c, a distance $r = 13.0\,cm$ away, is

$$\vec{\boldsymbol{r}}_1 = r\cos\alpha\ \hat{\boldsymbol{\imath}} + r\sin\alpha\ \hat{\boldsymbol{\jmath}}$$

Hence the unit vector that points from q_1 to c is

$$\hat{\boldsymbol{r}}_1 = \vec{\boldsymbol{r}}_1/r = \cos\alpha\ \hat{\boldsymbol{\imath}} + \sin\alpha\ \hat{\boldsymbol{\jmath}}$$

Evaluate

And the field due to q_1 at point c is

$$\vec{\boldsymbol{E}}_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r^2}\hat{\boldsymbol{r}}_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r^2}(\cos\alpha\ \hat{\boldsymbol{\imath}} + \sin\alpha\ \hat{\boldsymbol{\jmath}})$$

By symmetry the unit vector that points from q_2 to point c has the opposite x-component but the same y-component, so the field at c due to q_2 is

$$\vec{\boldsymbol{E}}_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r^2}\hat{\boldsymbol{r}}_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r^2}(-\cos\alpha\ \hat{\boldsymbol{\imath}} + \sin\alpha\ \hat{\boldsymbol{\jmath}})$$

Evaluate

Since $q_2 = -q_1$, the total field at c is:

$$\vec{\boldsymbol{E}}_c = \vec{\boldsymbol{E}}_1 + \vec{\boldsymbol{E}}_2$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r^2}(\cos\alpha\ \hat{\boldsymbol{\imath}} + \sin\alpha\ \hat{\boldsymbol{\jmath}}) + \frac{1}{4\pi\varepsilon_0}\frac{(-q_1)}{r^2}(-\cos\alpha\ \hat{\boldsymbol{\imath}} + \sin\alpha\ \hat{\boldsymbol{\jmath}})$$

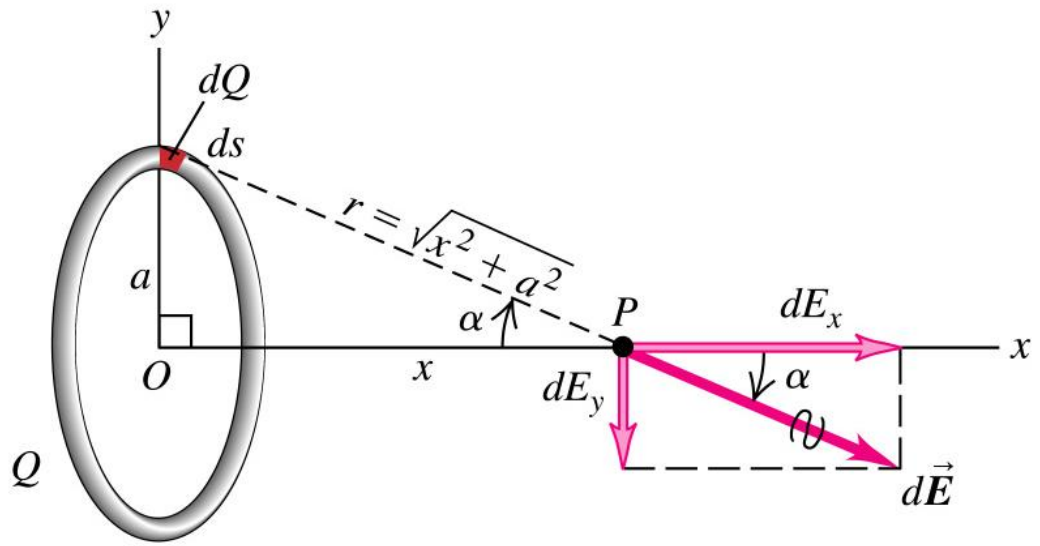
$$= \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r^2}(2\cos\alpha\ \hat{\boldsymbol{\imath}})$$

$$= (9.0\times10^9\ N\cdot m^2/C^2)\frac{12\times10^{-9}\ C}{(0.13\ m)^2}\left[2\left(\frac{5}{13}\right)\right]\hat{\boldsymbol{\imath}}$$

$$= (4.9\times10^3\ N/C)\ \hat{\boldsymbol{\imath}}$$

Example 21.10 Field of a ring of charge

A ring-shaped conductor with a radius a carries a total charge Q uniformly distributed around it Find the electric field at a point P that lies on the axis of the ring at a distance x from its center.



Identify and Set Up

This is the problem in the superposition of electric field. Each bit of charge around the ring produces an electric field at an arbitrary point on the x-axis. Our target variable is the total electric field at this point due to all such bits of charge.

Execute

As shown in the figure above, we imagine the ring divided into infinitesimal segments of length ds. Each segment has charge dQ and acts as point charge source of electric field.

Let $d\vec{E}$ be the electric field from one such segment; the net electric field at P is then the sum of all contributions $d\vec{E}$ from all the segments that make up the ring.

(This same technique works for any situation in which charge is distributed along a line or a curve.)

Execute

The calculation of $\vec{E}$ is greatly simplified because the field point P is on the symmetry axis of the ring.

Consider two segments at the top and bottom of the ring: the contributions $d\vec{E}$ to the field at P from these segments have the same x-component but opposite y-components.

Hence the total y-component of field due to this pair of segments is zero.

Execute

When we add up the contributions from all such pairs of segments, the total field $\vec{E}$ will have only a component along the ring's symmetry axis (the x-axis), with no component perpendicular to that axis (that is, no y-component or z-component).

So the field at P is described completely by its x-component E_x.

Execute

To calculate E_x, note that the square of the distance r from a ring segment to the point P is $r^2 = x^2 + a^2$.

Hence the magnitude of this segment's contribution $d\vec{E}$ to the electric field at P is

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dQ}{x^2 + a^2}$$

Execute

Using $\cos\alpha = \frac{x}{r} = \frac{x}{(x^2+a^2)^{1/2}}$, the x-component dE_x of this field is

$$\begin{aligned} dE_x &= dE\cos\alpha \\ &= \frac{1}{4\pi\varepsilon_0}\frac{dQ}{x^2+a^2}\frac{x}{\sqrt{x^2+a^2}} \\ &= \frac{1}{4\pi\varepsilon_0}\frac{xdQ}{(x^2+a^2)^{3/2}} \end{aligned}$$

Execute

To find the total x-component E_x of the field at P, We integrate this expression over all segments of the ring:

$$E_x = \int \frac{1}{4\pi\varepsilon_0} \frac{x dQ}{(x^2 + a^2)^{3/2}}$$

Since x does not vary as we move from point to point around the ring, all the factors on the right side except dQ are constant and can be taken outside the integral.

Execute

The integral of dQ is just the total charge Q, and we finally get

$$\vec{\boldsymbol{E}} = E_x \hat{\boldsymbol{\imath}} = \frac{1}{4\pi\varepsilon_0} \frac{Qx}{(x^2 + a^2)^{3/2}} \hat{\boldsymbol{\imath}} \qquad (21.8)$$

Evaluate

Our result for $\vec{E}$ shows that at the center of the ring ($x = 0$) the field is zero.

This makes sense; charges on opposite sides of the ring would push in opposite directions on a test charge at the center, and the vector sum of each such pair of forces is zero.

When the field point P is much farther from the ring than its size (that is, $x >> a$), the denominator in Eq. (21.8) becomes approximately equal to x^3 , and the expression becomes approximately

Evaluate

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{x^2}\hat{\imath}$$

In other words, when we are so far from the ring that its size a is negligible in comparison to the distance x, its field is the same as that of a point charge.

To an observer far from the ring, the ring would appear like a point, and the electric field reflects this.

Evaluate

In this example we used a symmetry argument to conclude that $\vec{E}$ had only an x-component at a point on the ring's axis of symmetry.

We'll use symmetry arguments many times in this and subsequent chapters.

Keep in mind, however, that such arguments can be used only in special cases.

Evaluate

At a point in the xy-plane that is not on the x-axis, the symmetry argument doesn't apply, and the field has in general both x- and y-components.

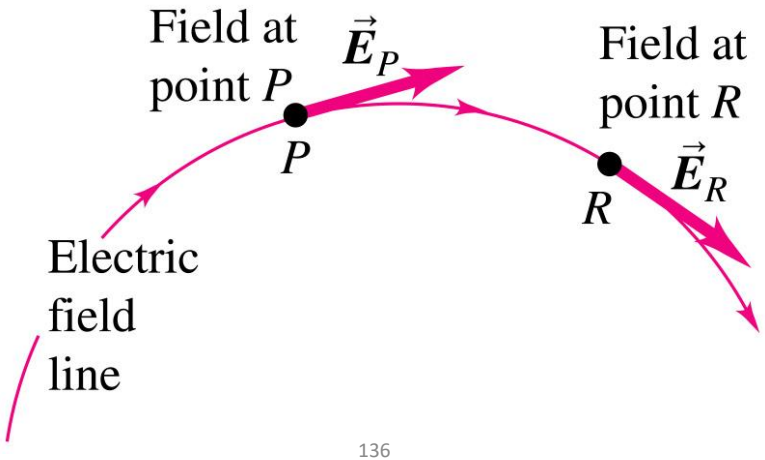
Chapter 21.6

Electric Field Lines

21.6 Electric Field Lines

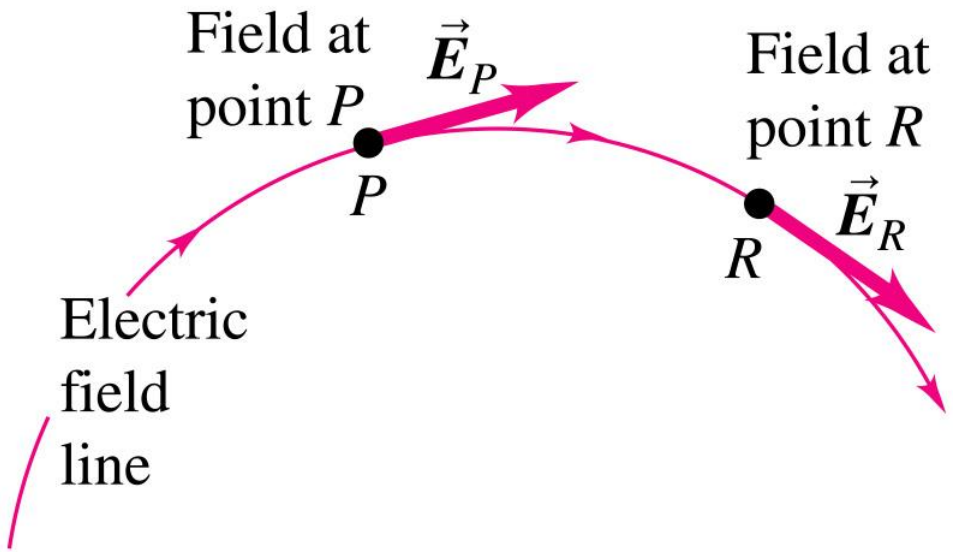
Electric Field Lines (visualize the electric field)

- An electric field line is an imaginary line or curve drawn through a region of space so that its tangent at any point is in the direction of the electric-field vector at that point.



Interpretation of Electric Field Lines

- The direction of the electric field vector at any point is tangent to the field line passing through that point and in the direction indicated by arrows on the field line.



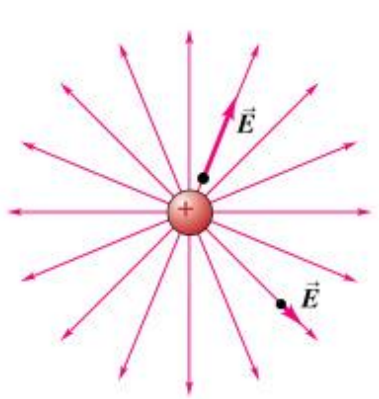
Interpretation of Electric Field Lines

- The electric field is strong where field lines are close together and weak where they are far apart.

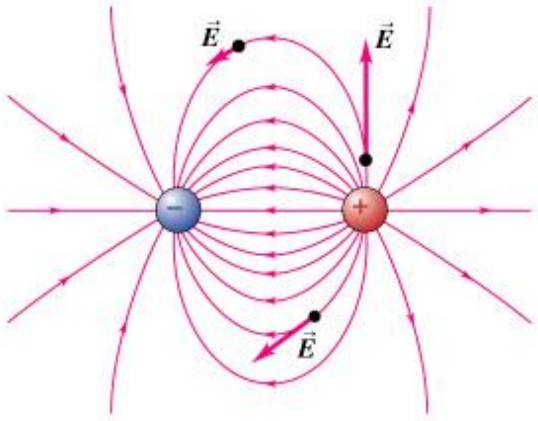
- More specifically, if you imagine small surface perpendicular to the field lines, the magnitude of the field is proportional to the number of lines that cross the surface divided by the area.

Rules for Sketching Field Lines

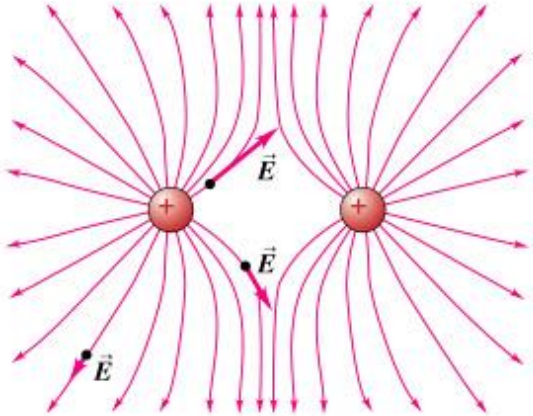
- Electric field lines can start only on positive charges and can end only on negative charges.



(a) A single positive charge (compare Figure 21.16)



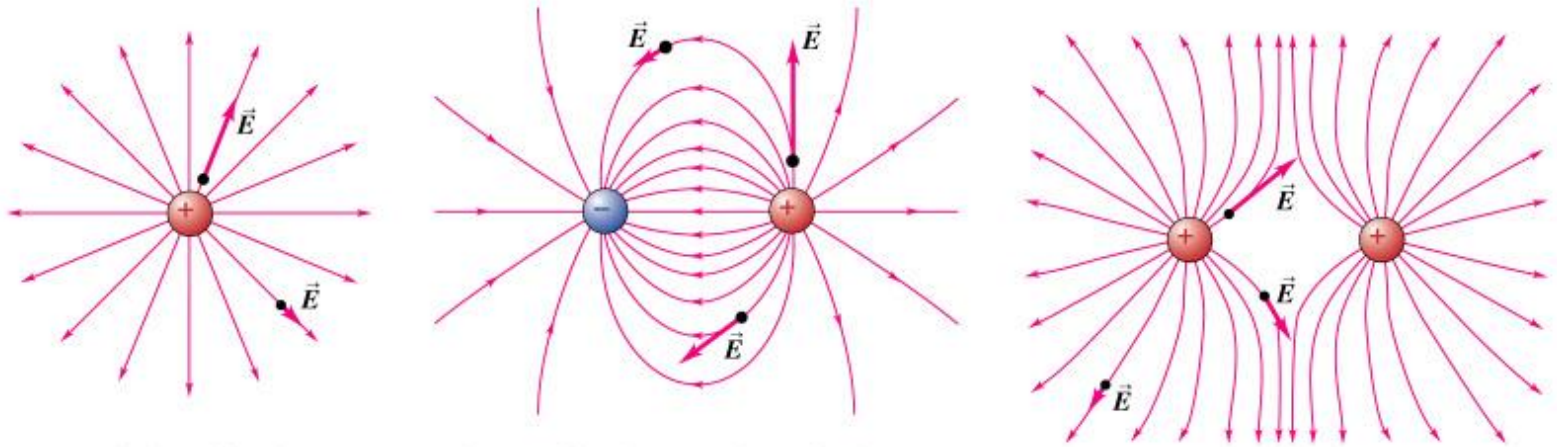
(b) A positive charge and a negative charge of equal magnitude (an electric dipole)



(c) Two equal positive charges

Rules for Sketching Field Lines

- The number of lines starting on a positive charge (or ending on a negative charge) is proportional to the magnitude of the charge.
- Field lines never cross.



(a) A single positive charge (compare Figure 21.16)

(b) A positive charge and a negative charge of equal magnitude (an electric dipole)

(c) Two equal positive charges

Chapter 21.7

Electric Dipoles

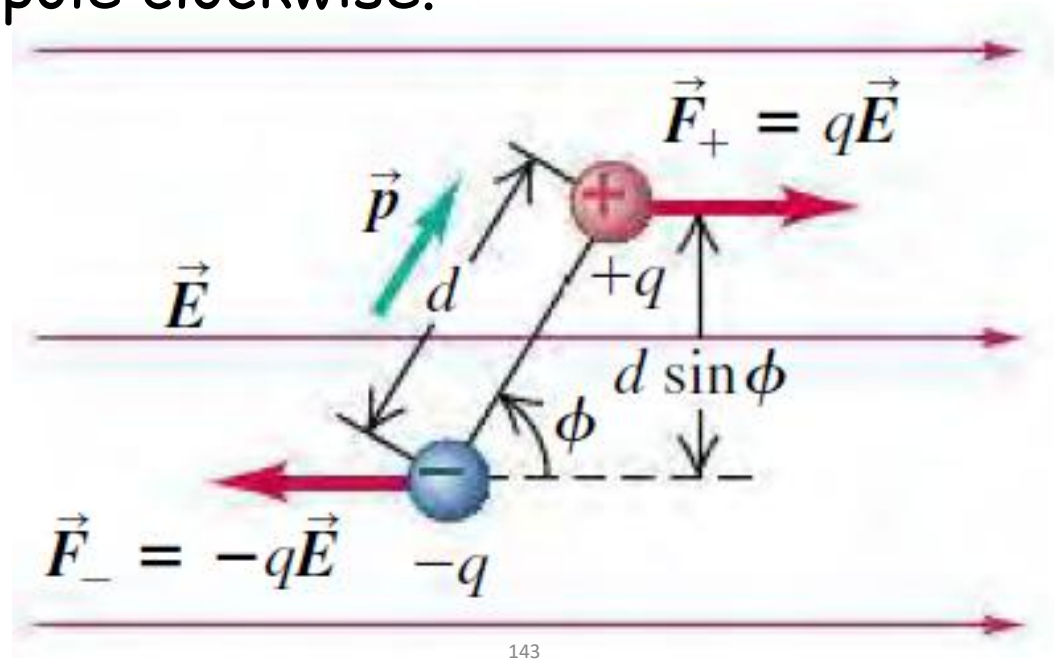
21.7 Electric Dipoles

- An electric dipole is a pair of point charges with equal magnitude and opposite sign (a positive charge q and a negative charge -q) separated by a distance d.

- The figure in the next slide shows the force and torque associated with a electric dipole.

21.7 Electric Dipoles

- The net force on this electric dipole is zero.
- But there is a torque directed into the page that tends to rotate the dipole clockwise.



21.7 Electric Dipoles

- The torque τ exerted by an uniform electric field is

$$\tau = pE \sin\phi$$

(magnitude of the torque on an electric dipole)

- This can be written as

$$\vec{\tau} = \vec{p} \times \vec{E}$$

(torque on an electric dipole, in vector form)

21.7 Electric Dipoles

- The work done by τ via the change in the direction of the electric dipole is thus

$$U = -\vec{\boldsymbol{p}} \cdot \vec{\boldsymbol{E}}$$

(potential energy for a dipole in an electric field)

Example 21.14 Force and torque on an electric dipole

Figure in the next slide shows an electric dipole in a uniform field with magnitude $5.0 \times 10^5\ N/C$ directed parallel to the plane of the figure. The charges are $\pm 1.6 \times 10^{-19}\ C$; both lie in the plane and are separated by $0.125\ nm = 0.125 \times 10^{-9}\ m$. (Both the charge magnitude and the distance are typical of molecular quantities.)

Find

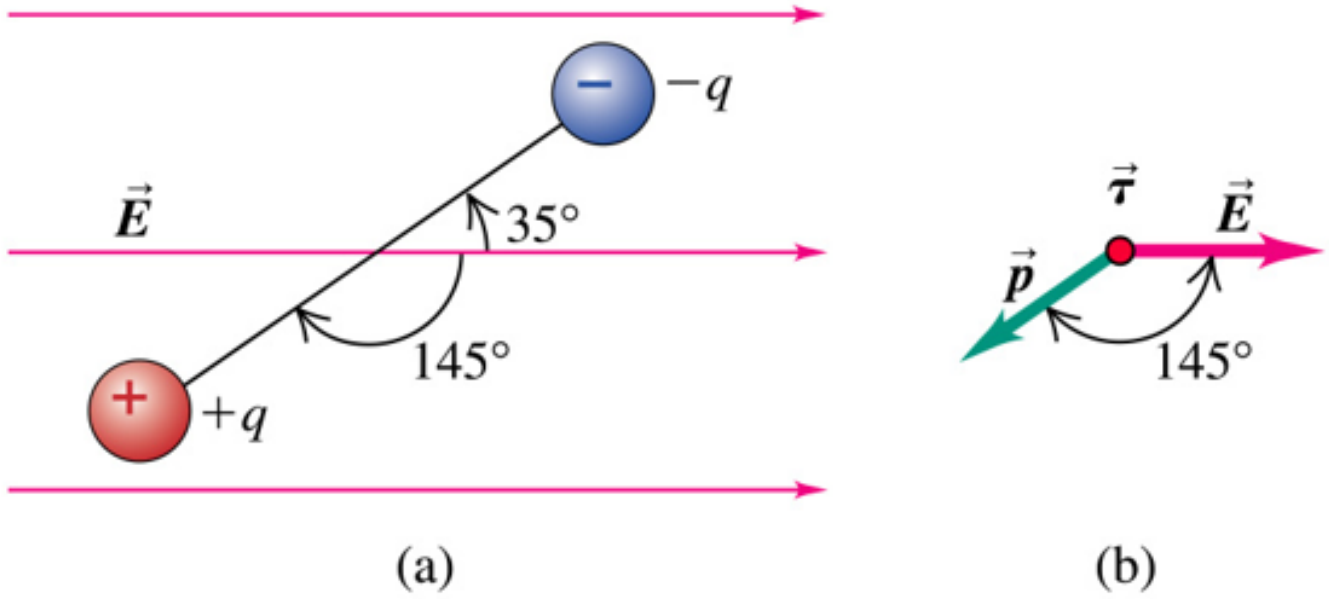
a) the net force exerted by the field on the dipole;

b) the magnitude and direction of the electric dipole moment;

Example 21.14

c) the magnitude and direction of the torque;

d) the potential energy of the system in the position shown.



Identify and Set Up

This problem uses the ideas of this section about an electric dipole placed in an electric field.

We use the relationship $\vec{F} = q\vec{E}$ for each point charge to find the force on the dipole as a whole. Equation (21.14) tells us the dipole moment, Eq. (21.16) tells us the torque on the dipole, and Eq. (21.18) tells us the potential energy of the system.

Execute

a) Since the field is uniform, the forces on the two charges are equal and opposite, and the total force is zero.

b) The magnitude p of the electric dipole moment $\vec{\boldsymbol{p}}$ is

$$\begin{aligned} p &= qd \\ &= (1.6 \times 10^{-19}\ C)(0.125 \times 10^{-9}\ m) \\ &= 2.0 \times 10^{-29}\ C \cdot m \end{aligned}$$

The direction of $\vec{\boldsymbol{p}}$ is from the negative to the positive charge, $145°$ clockwise from the electric-field direction.

Execute

c) The magnitude of the torque is

$$\begin{aligned}\tau &= pE\sin\phi \\ &= (2.0\times10^{-29}\ C)(5.0\times10^{5}\ N/C)(\sin 145°) \\ &= 5.7\times10^{-24}\ N\cdot m\end{aligned}$$

From the right hand rule for vector products, the direction of the torque $\vec{\tau}=\vec{p}\times\vec{E}$ is out of the page. This corresponds to a counterclockwise torque that tends to align $\vec{p}$ with $\vec{E}$.

Execute

d) The potential energy is

$$\begin{aligned} U &= -pE\cos\phi \\ &= -(2.0\times10^{-29}\ C)\left(5.0\times10^{5}\frac{N}{C}\right)(\cos 145°) \\ &= 8.2\times10^{-24}\ J \end{aligned}$$

Evaluate

The dipole moment, torque, and potential energy are all exceedingly small.

Don't be surprised by this result: remember that we are looking at a single molecule, which is a very small object indeed!

Example 21.15 Field of an electric dipole, revisited

In Fig. 21.31 an electric dipole is centered at the origin, with $\vec{p}$ in the direction of the $+y$-axis.

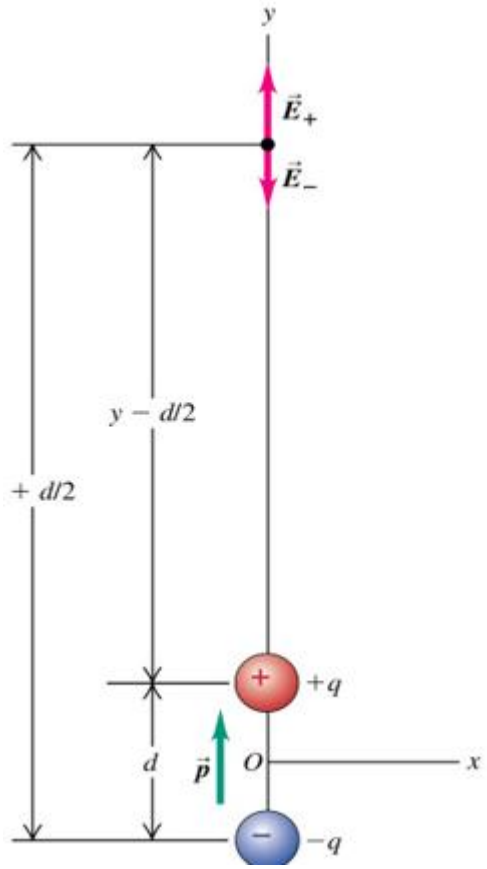
Derive an approximate expression for the electric field at a point on the y-axis for which y is much larger than d.

Use the binomial expansion of $(1+x)^n$, that is, $(1+x)^n \cong 1+nx+n(n-1)x^2/2+\cdots$, for the case $|x|<1$. (This problem illustrates a useful calculational technique.)

Identify

We use the principle of superposition:

the total electric field is the vector sum of the field produced by the positive charge and the field produced by the negative charge.



Set up

At the field point shown in the figure above, the field of the positive charge has a positive (upward) y-component and the field of the negative charge has a negative (downward) y-component.

We add these components to find the total field, then apply the approximation that y is much greater than d.

Execute

The total y −component E_y of electric field from the two charges is

$$E_y = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(y-d/2)^2} - \frac{1}{(y+d/2)^2}\right]$$
$$= \frac{q}{4\pi\varepsilon_0 y^2}\left[\left(1-\frac{d}{2y}\right)^{-2} - \left(1+\frac{d}{2y}\right)^{-2}\right]$$

We used this exact approach in Example 21.9 (Section 21.5). Now comes the approximation.

Execute

When y is much greater than d, that is, when we are far away from the dipole compared to its size, the quantity $d/2y$ is much smaller than 1.

With $n = -2$ and $d/2y$ playing the role of x in the binomial expansion, we keep only the first two terms.

The terms we discard are much smaller than those we keep, and we have

Execute

$$(1-\frac{d}{2y})^{-2}\cong 1+\frac{d}{y} \quad and\ (1+\frac{d}{2y})^{-2}\cong 1-\frac{d}{y}$$

Hence E_y is approximately given by

$$E \cong \frac{q}{4\pi\varepsilon_0 y^2}\left[1+\frac{d}{y}-\left(1-\frac{d}{y}\right)\right]$$
$$= \frac{qd}{2\pi\varepsilon_0 y^3} = \frac{p}{2\pi\varepsilon_0 y^3}$$

Evaluate

An alternative route to this expression is to put the fractions in the E_y expression over a common denominator and combine, then approximate the denominator $(y - d/s)^2(y + d/s)^2$ as y^2.

We leave the details as a problem.

For points P off the coordinate axes, the expressions are more complicated, but at all points far away from the dipole (in any direction) the field drops off as $1/r^3$.

Evaluate

We can compare this with the $1/r^2$ behavior of a point charge, the $1/r$ behavior of a long line charge, and the independence of r for a large sheet of charge.

There are charge distributions for which the field drops off even more quickly.

An electric quadrupole consists of two equal dipoles with opposite orientation, separated by a small distance. The field of a quadrupole at large distances drops off as $1/r^4$.

Summary

Electric charge, conductors, and insulators

The fundamental quantity in electrostatics is electric charge.

There are two kinds of charge, positive and negative.

Charges of the same sign repel each other, charges of opposite sign attract.

Electric charge, conductors, and insulators

Charges is conserved: the total charge in an isolated system is constant.

All ordinary matter is made of protons, neutrons, and electrons.

The positive protons and electrically neutral neutrons in the nucleus of an atom are bound together by the nuclear force;

The negative electrons surround the nucleus at a distance much great than the nuclear size.

Electric charge, conductors, and insulators

Electric interactions are chiefly responsible for the structure of atoms, molecules, and solids.

Conductors are materials in which charge moves easily: in insulators, charge does not move easily. Most metals are good conductors; most nonmetals are insulators.

Coulomb's law

For charges q_1 and q_2 separated by a distance r, the magnitude of the electric force on either charge is proportional to the product $q_1 q_2$ and inversely proportional to r^2.

The force on each charge is along the line joining the two charges: repulsive if q_1 and $q_2$have the same sign, attractive if they have opposite signs.

In SI units the unit of electric charges is the coulomb, abbreviated C.

Coulomb's law

When two or more charges each exert a force on a charge, the total force on that charge is the vector sum of the forces exerted by the individual charges.

$$F = \frac{1}{4\pi\varepsilon_0}\frac{|q_1 q_2|}{r^2}$$

$$\frac{1}{4\pi\varepsilon_0} = 8.988 \times 10^9\ N \cdot m^2/C^2$$

Electric field

Electric field $\vec{E}$, a vector quantity, is the force per unit charge exerted on a test charge at any point.

The electric field produced by a point charge is directed radially away from or toward the charge.

$$\vec{E} = \frac{\vec{F}_0}{q_0}$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

Superposition of electric fields

The electric field $\vec{E}$ of any combination of charges is the vector sum of the fields caused by the individual charges.

To calculate the electric field caused by a continuous distribution of charge, divide the distribution into small elements, calculate the field caused by each element, and then carry out the vector sum, usually by integrating.

Superposition of electric fields

Charge distributions are described linear charge density λ, surface charge density σ, and volume charge density ρ.

Electric field lines

Field lines provide a graphical representation of electric fields.

At any point on a field line, the tangent to the line is in the direction of $\vec{E}$ at that point.

The number of lines per unit area (perpendicular to their direction) is proportional to the magnitude of $\vec{E}$ at the point.

Electric dipoles

An electric dipole is a pair of electric charges of equal magnitude q but opposite sign, separated by a distance d.

The electric dipole moment $\vec{p}$ has magnitude $p = qd$. The direction of $\vec{p}$ is from negative toward positive charge.

An electric dipole in an electric field $\vec{E}$ experiences a torque $\vec{\tau}$ equal to the vector product of $\vec{p}$ and $\vec{E}$.

Electric dipoles

The magnitude of the torque depends on the angle ϕ between $\vec{\boldsymbol{p}}$ and $\vec{\boldsymbol{E}}$. The potential energy U for an electric dipole in an electric field also depends on the relative orientation of $\vec{\boldsymbol{p}}$ and $\vec{\boldsymbol{E}}$.

$$\tau = pE\sin\phi$$

$$\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{p}} \times \vec{\boldsymbol{E}}$$

$$U = -\vec{\boldsymbol{p}} \cdot \vec{\boldsymbol{E}}$$

Homework

21.33; 21.44

Hand in your homework every Tuesday!

Chapter 21 Electric Charge and Electric Field

21.33 An electron is projected with an initial speed $v_0 = 1.60 \times 10^6\, m/s$ into the uniform field between the parallel plates in the figure in this problem. Assuming that the field between the plates is uniform and directed vertically downward, and that the field outside the plates is zero. The electron enters the field at a point midway between the plates.

(a) If the electron just misses the upper plate as it emerges from the field, find the magnitude of the electric field.

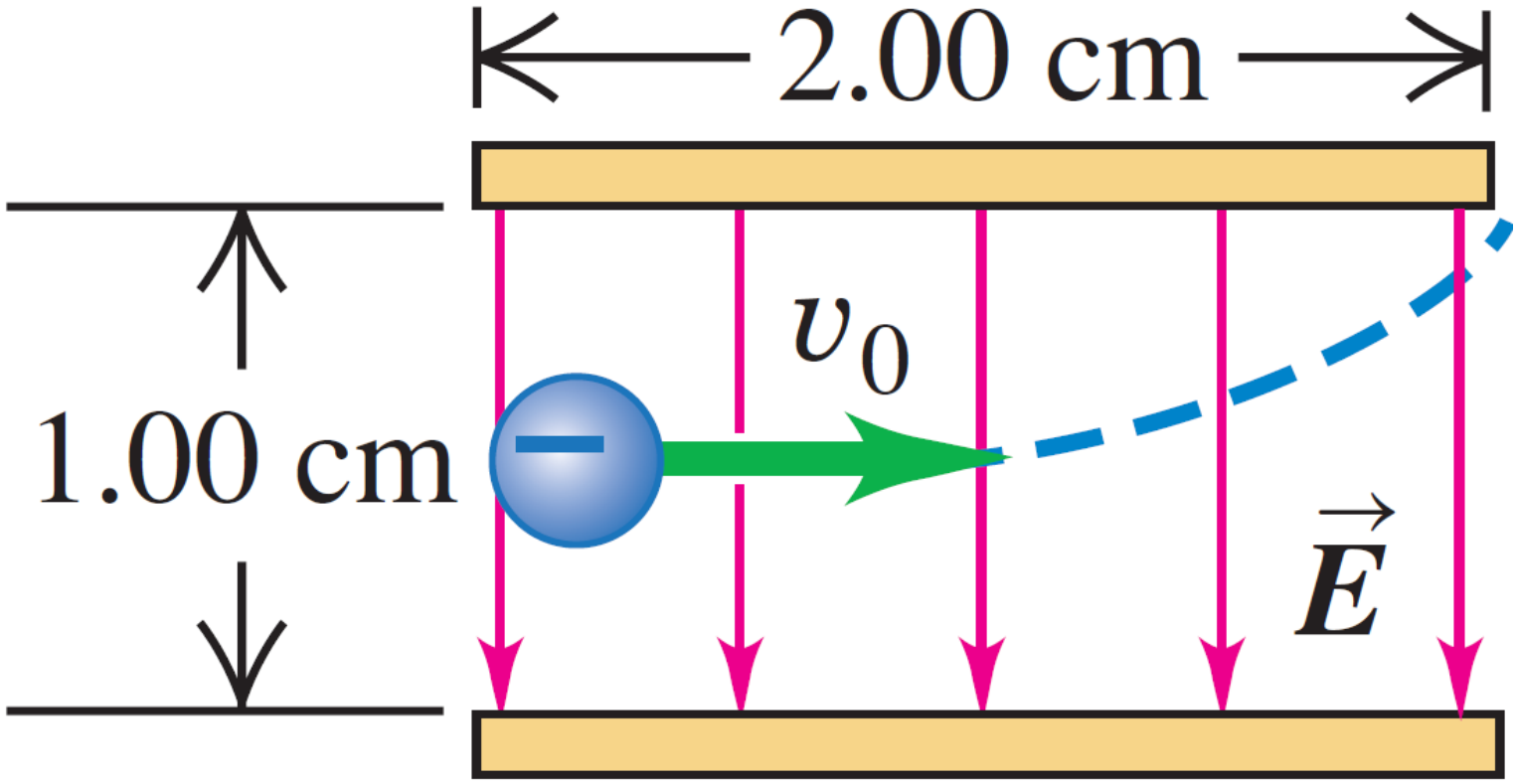
Chapter 21 Electric Charge and Electric Field

b) Suppose that in the figure the electron is replaced by a proton with the same initial speed v_0. Would the proton hit one of the plates? If the proton would not hit one of the plates, what would be the magnitude and direction of its vertical displacement as it exits the region between the plates?

c) Compare the paths traveled by the electron and the proton and explain the difference.

d) Discuss whether it is reasonable to ignore the effects of gravity for each particle.

Chapter 21 Electric Charge and Electric Field



Chapter 21 Electric Charge and Electric Field

21.44 The two charges q_1 and q_2 shown in the figure have equal magnitudes.

What is the direction of the net electric field due to these two charges at point A (midway between the charges), B, and C if

(a) both charges are positive,

(b) both charges are negative,

(c) q_1 is positive and q_2 is negative.

Chapter 21 Electric Charge and Electric Field

